

Deposit Pricing, Bank Funding Models, and the Incidence of Reserve Requirements

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Abstract

This paper compares the relationship between bank funding structure and reserve requirements across deposit-pricing regimes. Using quarterly data for five large state-owned and nine national joint-stock banks in China over 2008–2024, I measure predetermined exposure by standardized 2008–2011 time-deposit shares. Market funding comprises bonds payable and repo liabilities relative to total liabilities. In the baseline specification, the fitted market-funding gap between joint-stock banks one standard deviation apart in exposure widens by 1.144 percentage points for each percentage point of the reserve ratio during the regulated-pricing period. The corresponding gradient is -0.024 after formal ceiling removal, giving a regime difference of -1.168 and a joint-stock-minus-large-bank contrast of -1.388 . The cross-type contrast remains negative across alternative predetermined exposures, policy-variable constructions, and individual-bank exclusions, although small-cluster inference is less precise than conventional clustering. Limited common support and sensitivity to exposure-related trends prevent attributing the difference to deposit-price liberalization. The findings document a conditional funding pattern and underscore the importance of checking stability when interpreting exposure-based estimates across policy environments.

JEL codes: G21, G28, E43, E52.

Keywords: bank funding; deposit regulation; reserve requirements; liability management; financial liberalization; China.

1 Introduction

Bank funding structure is often used to organize heterogeneous responses to reserve requirements. Applying an exposure-based estimate across policy environments, however, requires the relationship between that characteristic and bank funding to remain stable. This condition matters when deposit-price regulation changes alongside reserve policy. Reserve requirements remain an active instrument in emerging markets, where bank responses can depend on funding structure and liquidity (Brei and Moreno, 2019; Brei et al., 2026). This paper asks how stable exposure-based bank-funding gradients are across deposit-pricing and reserve-policy environments.

Reserve requirements affect the relative cost of reservable deposits and other liabilities. Banks can adjust through deposit prices, quantities, product composition, and market funding, but a binding deposit-rate ceiling restricts the price margin. China relaxed its formal deposit-rate ceilings between 2012 and 2015, while reserve requirement ratios (RRRs) subsequently declined substantially (People’s Bank of China, 2012, 2014, 2015, 2016; Tan et al., 2016). Formal ceiling removal did not eliminate all managed deposit pricing. Nationwide banks also entered this transition with different funding structures: joint-stock banks relied more on market and interbank liabilities than large state-owned banks. These differences motivate a comparison across funding models without equating bank ownership with a particular mechanism.

I use quarterly data for five large state-owned and nine national joint-stock banks over 2008–2024. Predetermined exposure is the standardized bank-level mean of time deposits relative to customer deposits during 2008–2011. Market funding comprises bonds payable and repurchase-agreement (repo) liabilities relative to total liabilities. The specification compares conditional levels gradients across pricing regimes, with bank and bank-type-by-quarter fixed effects. The latter absorb common quarterly level shocks within each bank type; the comparison comes from interactions of the policy variables with predetermined exposure.

In the baseline specification, two joint-stock banks one standard deviation apart in exposure have a fitted market-funding gap that widens by 1.144 percentage points of liabilities for each percentage point of the RRR during the regulated-pricing period. The corresponding later-regime gradient is -0.024 , giving a regime difference of -1.168 . The later point estimate is close to zero, but its uncertainty does not establish economic equivalence to zero.

The estimated regime difference for large state-owned banks is smaller, at 0.219, producing a joint-stock-minus-large-bank contrast of -1.388 . This comparison describes the baseline pattern across two funding models. It is not an ownership treatment effect, and the illustrative liability-choice model does not deliver a bank-type ranking without additional parameter restrictions. Common-sample liability outcomes also do not establish a one-for-one substitution from market funding into customer deposits.

The paper relates to research on deposit pricing and deposit franchises. Market power, switching frictions, and franchise value shape deposit-rate adjustment and monetary transmission (Berger and Hannan, 1989; Hannan and Berger, 1991; Drechsler et al., 2017, 2021; Egan et al., 2017; Wang et al., 2022). Evidence on U.S. Regulation Q and negative policy rates illustrates how constraints

on deposit prices can affect bank credit and the relevance of deposit dependence (Mertens, 2008; Koch, 2015; Heider et al., 2019). For China, Chen et al. (2023) study how deposit regulation changes retail and wholesale funding responses to policy rates; Chen et al. (2025) examine the associated transmission–systemic-risk trade-off. The object here is a conditional relationship with the reserve-ratio level, comparing an exposure measured before the pricing transition across historical environments.

The paper also connects to liability substitution and reserve-policy transmission. Bank liability structure and funding-market conditions are related to funding costs and risk (Craig and Dinger, 2013; Hahm et al., 2013; Acharya and Mora, 2015; Gerlach et al., 2018; Enkhbold, 2026), while deposit competition and liability regulation can redirect Chinese bank funding (Acharya et al., 2025; Liu et al., 2025). Catão and te Kaat (2021) examine bank-liability composition following capital-account liberalization, and Andreasen and Nuguer (2025) study reserve requirements and capital inflows in a partially dollarized economy. Together with evidence on heterogeneous reserve-requirement responses in China (Fungáčová et al., 2016), these studies motivate examining the funding characteristic used to organize heterogeneity. The present contribution is a bank-panel comparison of that characteristic’s fitted RRR gradient across deposit-pricing regimes, with explicit checks on policy support and exposure-related trends.

These comparisons are conditional historical associations. Reserve-ratio support shifts across periods, and exposure-related trends materially alter the within-type estimates and, when allowed to differ by bank type, the cross-type contrast. Small-cluster procedures also give weaker precision for the main cross-type comparison than conventional clustering. The support and sensitivity analyses therefore bound the interpretation of the baseline pattern rather than identify a separate causal effect of deposit-price liberalization. The results motivate checking stability when interpreting exposure-based estimates across environments; they do not establish structural non-portability, international spillovers, or an aggregate monetary multiplier.

Section 2 describes the institutions and economic framework. Section 3 defines the data and estimands, and Section 4 presents the baseline gradients. Section 5 examines policy support, trends, and timing. Sections 6 and 7 report liability evidence and further sensitivity analyses. Section 8 concludes.

2 Institutional Setting and Liability-Management Framework

2.1 Reserve requirements and deposit pricing

The reserve requirement ratio is the share of reservable liabilities that a bank must hold as reserves. When these holdings earn less than alternative assets, reserve requirements change the relative cost of deposit and nondeposit funding. Banks may adjust deposit prices, quantities, maturity and product composition, market liabilities, liquid assets, or balance-sheet scale. Reserve requirements have remained an important policy instrument in China (Ma et al., 2013; Fungáčová et al., 2016) and other emerging economies (Agénor et al., 2018; Brei and Moreno, 2019; Brei et al., 2026).

Other central-bank balance-sheet operations can also affect lending: Yun (2020) studies foreign-reserve accumulation and sterilization in Korea, a distinct policy object from the statutory reserve requirement considered here.

A binding deposit-rate ceiling limits the price offered on a regulated product. Market liabilities can then provide an alternative funding margin when deposit procurement is constrained. This motivates comparing exposure gradients across pricing regimes, but separating pricing from other concurrent changes requires more than a common national chronology. In particular, the policy values observed in each regime may differ.

2.2 Bank funding models

Before 2012, joint-stock banks had lower customer-deposit shares and higher market- and interbank-funding shares than large state-owned banks. Their deposits-to-liabilities ratio averaged 77.5 percent, compared with 84.3 percent for large banks; bonds plus repos accounted for 5.5 percent versus 1.6 percent; and interbank liabilities accounted for 17.6 percent versus 11.5 percent. Joint-stock banks were also smaller. Table 1 reports these descriptive differences.

Table 1: Pre-reform funding models

Bank characteristic	Large banks	Joint-stock banks	JSB minus large	Large obs.	JSB obs.
Log total assets	29.729	27.910	-1.819	68	113
Customer deposits / liabilities (%)	84.308	77.458	-6.851	68	113
Time deposits / customer deposits (%)	45.884	46.758	0.874	45	64
Bonds plus repos / liabilities (%)	1.586	5.543	3.957	68	113
Interbank liabilities / liabilities (%)	11.493	17.557	6.064	68	113
Banks	5	8			

Notes: Entries are unweighted means of bank-level averages over pre-2012 observations with the market-funding outcome; 181 bank-quarters and thirteen banks. Zheshang has no outcome in this window. Time-deposit shares use their own available reports. Repos appear in both bonds-plus-repos funding and reported interbank liabilities, so those rows overlap and are not additive. Ratios are percentage points; differences are descriptive.

The table uses 181 pre-2012 bank-quarters with the market-funding outcome for thirteen banks. Zheshang has no market-funding observation in that window, although its separate time-deposit disclosures permit construction of pre-reform exposure. Rows use their available observations. Time-deposit shares average 45.9 percent for large banks and 46.8 percent for joint-stock banks, so the group comparison does not simply reflect different mean exposure. Bank type also combines ownership, size, branch networks, state support, and funding access. It is a descriptive funding-model comparison.

2.3 Deposit-pricing transition and the RRR environment

The People’s Bank of China (PBoC) relaxed deposit ceilings through six direct ceiling or product-scope changes from June 2012 to October 2015 (People’s Bank of China, 2012, 2014, 2015, 2016; Tan

et al., 2016). The primary regime index records this sequence and excludes the separate reform of large-denomination certificates sold to customers. Completion of formal ceiling removal is distinct from unrestricted pricing. For example, the PBoC’s 2021Q3 monetary policy report describes an adjustment to deposit-rate self-regulatory ceilings (People’s Bank of China, 2021). The later period is therefore called the post-ceiling-removal regime; the model’s limiting case of full flexibility is not a literal description of post-2015 China.

The transition coincided with changes in the operating framework (Fernald et al., 2014), lower RRR levels, deposit insurance, interbank negotiable certificates of deposit (NCDs), and other financial reforms (State Council of the People’s Republic of China, 2015; Market Interest Rate Pricing Self-Discipline Mechanism, 2015; Chen et al., 2018). Interbank NCDs are distinct from large-denomination certificates sold to customers. Figure 1 presents the pricing index and bank-type RRR paths together. Appendix A documents the chronology and concurrent changes.

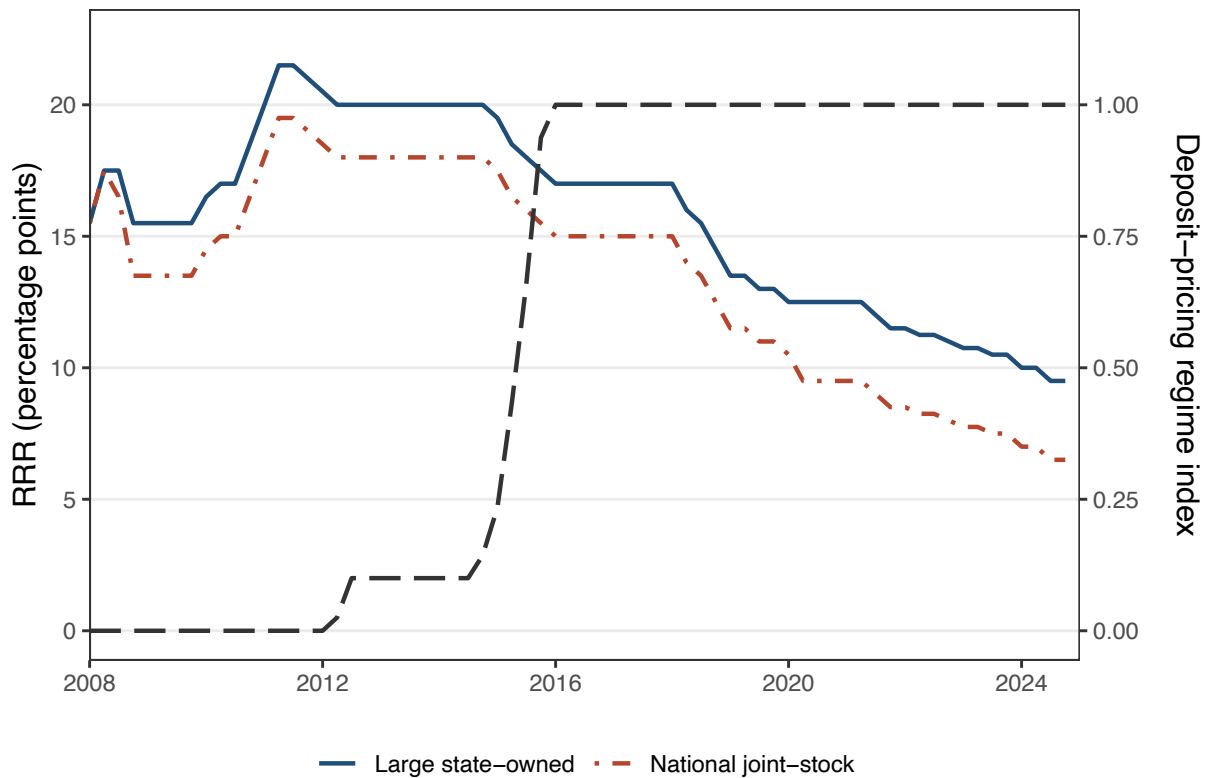


Figure 1: Deposit-pricing transition and bank-type RRR paths

Notes: The long-dashed line reports the primary pure deposit-ceiling regime index on the right axis. The solid and dot-dash lines report quarter-end RRR paths for large state-owned and national joint-stock banks, respectively. The deposit-pricing transition and a decline in RRR levels occur over the same historical period; the full-sample interaction is therefore interpreted as a regime-associated gradient rather than a causal effect of deposit-rate liberalization.

2.4 Economic framework

Consider a bank of type j choosing customer deposits d and market funding w to meet a fixed gross local funding target F , so $d + w = F$. Other balance-sheet items are held fixed; F is not lending net of required reserves. Let $\alpha \in [0, 1)$ be the reserve rate, $\theta \in [0, 1]$ a raw time-deposit share, and $L \in [0, 1]$ pricing flexibility. A benchmark with constant deposit-cost curvature has costs

$$\begin{aligned} C_j^D(d) &= [\mu_j + \alpha\{\rho_{0j} + \rho_{1j}\theta(1 - L)\}]d + \frac{\kappa_j}{2}d^2, \\ C_j^W(w) &= \xi_j w + \frac{\nu_j}{2}w^2. \end{aligned} \tag{1}$$

Assume $\kappa_j, \nu_j > 0$, $\rho_{0j} \geq 0$, and $\rho_{1j} > 0$. The reserve-cost wedge is then nonnegative. At an interior optimum,

$$\frac{\partial^2 w_j^*}{\partial \alpha \partial \theta} = \frac{\rho_{1j}(1 - L)}{\kappa_j + \nu_j}. \tag{2}$$

Thus a pricing-related reserve wedge can generate a positive exposure gradient that attenuates with flexibility. Appendix D allows exposure-dependent curvature; attenuation then requires its stated inequality. Neither version is structurally estimated or determines the ranking of the two bank types without restrictions on their parameters.

The model uses funding levels and raw exposure, whereas the regressions use percentage shares and standardized exposure. With fixed total liabilities T , $MF = 100w/T$, $RRR = 100\alpha$, and $E = (\theta - \mu_0)/\sigma_0$, the empirical-unit derivative equals $(\sigma_0/T)\partial^2 w/(\partial \alpha \partial \theta)$. This preserves the sign but does not identify the regression magnitude with a structural parameter. The framework illustrates a possible funding margin, consistent with work on deposit franchises and wholesale funding (Drechsler et al., 2017, 2021; Wang et al., 2022; Enkhbold, 2026).

2.5 Other changes in bank funding markets

Interbank NCDs changed the funding opportunities available to banks during the transition. Legal eligibility, first observed issuance, and realized use are different objects: eligibility affects the feasible set, issuance is an endogenous adoption decision, and subsequent balances are funding choices. The matched-sample NCD exercises describe their association with the historical gradient, rather than treating them as exogenous controls that isolate deposit pricing. Related evidence documents how deposit competition and liability regulation can shift Chinese bank funding toward substitutes (Chen et al., 2018; Acharya et al., 2025; Liu et al., 2025).

3 Data and Empirical Design

3.1 Sample and market-funding outcome

The panel covers fourteen nationwide banks over 2008Q1–2024Q4: five large state-owned commercial banks and nine national joint-stock commercial banks. Eleven were already publicly listed before the sample begins. Agricultural Bank and Everbright listed in 2010; Zheshang first listed in

Hong Kong in March 2016 and in Shanghai on November 26, 2019 (China Zheshang Bank, 2019). Earlier coverage for these three banks is less frequent, consistent with changing disclosure frequency, without attributing every missing report to listing status. Appendix C documents coverage and source reconciliation.

The baseline contains 880 bank-quarter observations and all 68 calendar quarters. The outcome is

$$MF_{it} = 100 \frac{\text{bonds payable}_{it} + \text{repo liabilities}_{it}}{\text{total liabilities}_{it}}. \quad (3)$$

It combines two separately observed nondeposit funding categories. A broader noncustomer measure adds bonds payable to reported interbank liabilities, which already include repos. Adding interbank liabilities to MF would count repos twice. Component outcomes are analyzed separately rather than entered as contemporaneous controls for their aggregate. A supplementary annual sample contains 156 bank-years for ten banks with deposit-rate disclosures; full-year rates are not interpolated into quarterly observations.

3.2 Predetermined time-deposit exposure

Let $\mathcal{T}_i^0$ be the observed report periods during 2008Q1–2011Q4 with valid corporate and personal time-deposit components and customer deposits. Define raw and standardized exposure separately:

$$s_i = \frac{1}{|\mathcal{T}_i^0|} \sum_{t \in \mathcal{T}_i^0} \frac{TD_{it}^{corp} + TD_{it}^{pers}}{D_{it}}, \quad E_i = \frac{s_i - \mu_0}{\sigma_0}. \quad (4)$$

The regressions use E_i . The fixed scaling population comprises 41 banks with a finite pre-2012 bank-level mean in the upstream exposure file, weighted equally across banks. Its mean and sample standard deviation are approximately 0.4241 and 0.1217. The scale is held fixed when source corrections update s_i for estimation banks; Appendix B and the variable dictionary provide the roster and full-precision constants.

Time-deposit exposure uses six to sixteen observed reports per bank and no information after 2011Q4. Its measurement sample differs from the market-funding sample because the liability components needed for the two objects differ. Predetermined exposure is a cross-sectional characteristic, not policy assignment. Appendix Figure B.1 shows its distribution, and Table G.2 separates alternative windows, within-type standardization, and residualization.

3.3 Policy variables

The primary pricing index gives each of the six direct deposit-ceiling reforms a daily increment of 1/6 from its effective date and averages the daily values within each quarter. Thus $L_t = 0$ is the regulated regime, $0 < L_t < 1$ the transition, and $L_t = 1$ the completed formal-removal regime. October 2015 is the endpoint event; 2016Q1 is the first full quarter with $L_t = 1$. The secondary index uses cumulative daily values 0.2, 0.4, 0.6, 0.8, 0.9, 1.0 on the same dates. These weights provide an alternative chronology that places less weight on the two final scope removals; they are not

estimated economic effects. Both indices exclude the large-customer-CD reform. Appendix A gives the exact construction.

RRR is measured in percentage points. The large-bank and small/medium-bank benchmark series are assigned to large state-owned and joint-stock banks, respectively. The retained series originate from the project’s monthly workbook paths. Each month-end value is assigned to the calendar days in that month; quarter-end and quarter-start measures take the last and first days, and the average measure takes the mean over all days. This is a group benchmark with monthly timing, not each consolidated bank’s exact effective reserve requirement. Bank-specific concessions are not separately measured. The separate effective-date event file is documented in the dictionary but is not substituted for this benchmark. This distinction limits interpretations of within-quarter policy timing.

3.4 Specification and estimands

Let $g(i) \in \{G, J\}$ denote large state-owned and joint-stock banks. The pooled specification is

$$Y_{it} = \alpha_i + \lambda_{g(i),t} + \sum_{g \in \{G, J\}} \mathbf{1}\{g(i) = g\} [\beta_{1g} RRR_{g,t} E_i + \beta_{2g} L_t E_i + \beta_{3g} RRR_{g,t} L_t E_i] + \varepsilon_{it}. \quad (5)$$

Bank fixed effects absorb permanent bank differences. Bank-type-by-quarter fixed effects absorb quarterly level shocks common to each type, including the policy-variable levels. The central estimand is

$$g_g(L) = \frac{\partial^2 \widehat{MF}}{\partial RRR \partial E} = \beta_{1g} + \beta_{3g} L. \quad (6)$$

The regulated gradient is β_{1g} , the later gradient is $\beta_{1g} + \beta_{3g}$, and their difference is β_{3g} . The cross-type contrast is $\beta_{3J} - \beta_{3G}$. These are changes in a fitted funding gap between banks as the RRR level differs, not the entire funding gap or a dynamic response to an RRR shock.

3.5 Identification, support, and inference

The exposure gradient uses predetermined cross-bank variation interacted with the bank-type policy path. Interpreting differences across regimes requires a comparable RRR–exposure relationship despite shifts in policy support and other exposure-related characteristics. Fixed effects do not absorb shocks whose incidence within a type is systematically related to exposure. Section 5 therefore examines support restrictions, exposure-related trends, and timing. The baseline remains a conditional historical association.

Conventional uncertainty uses bank-clustered standard errors with a finite-sample correction and a t reference with thirteen degrees of freedom. Principal estimates also report CR2, a leverage-adjusted cluster-robust covariance estimator with Satterthwaite degrees of freedom, and restricted wild-cluster bootstrap tests (Cameron et al., 2008; MacKinnon and Webb, 2017; Pustejovsky and Tipton, 2018; MacKinnon et al., 2023). The bootstrap imposes the null, resamples bank-level

weights, and studentizes the statistic (WCR11). Rademacher tests enumerate all bank sign patterns; Webb tests use 9,999 draws. Two-sided tail counts include numerical ties. Different methods attach uncertainty to the same estimates and need not rank their p -values uniformly. Appendix B.1 gives the algorithm and finite-support details.

4 Market-Funding Exposure Gradients across Regimes

4.1 Joint-stock banks

Table 2 reports a regulated-period JSB gradient of 1.144. For two banks one upstream standard deviation apart in exposure, the fitted market-funding gap is 1.144 percentage points larger at an RRR one percentage point higher. The later point estimate is -0.024 , giving a difference of -1.168 . The CR2 p -value for that difference is 0.066. Figure 2 displays the same decomposition.

Table 2: Market-funding exposure gradients across regimes

Estimand	Estimate (SE)	p : CR1	p : CR2	p : WCR-R	p : WCR-W
JSB regulated gradient	1.144 (0.473)	0.031	0.095	0.027	0.035
JSB regime difference	-1.168 (0.435)	0.019	0.066	0.012	0.015
JSB later gradient	-0.024 (0.165)	0.884	0.888	0.895	0.879
Large-bank regime difference	0.219 (0.134)	0.126	0.265	0.312	0.296
JSB minus large-bank difference	-1.388 (0.455)	0.009	0.071	0.075	0.083

Notes: Market funding is 100 times bonds plus repos divided by total liabilities. Gradients are percentage points per RRR percentage point and upstream exposure standard deviation. All five rows use the same 880 observations, fourteen banks (nine JSB, five large), and bank and bank-type-by-quarter fixed effects. CR1 is the conventional bank-clustered covariance with the HC1 and cluster finite-sample factors and a t_{G-1} reference. CR2 uses leverage adjustment and Satterthwaite degrees of freedom. WCR-R and WCR-W are null-imposed, studentized WCR11 tests using Rademacher and Webb bank weights. Rademacher enumerates $2^{14} = 16,384$ patterns; Webb uses 9,999 draws. Two-sided tail counts include numerical ties within $10^{-8} \max(1, |t|)$. Appendix B.1 documents the tests.

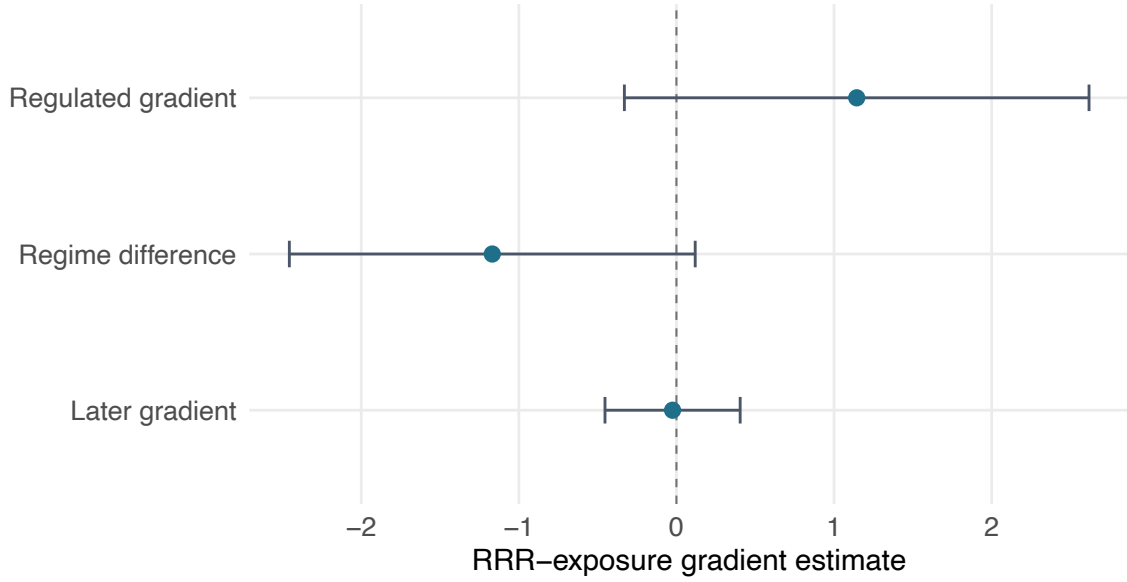


Figure 2: Joint-stock banks: market-funding slope decomposition
Notes: The figure reports the regulated-regime RRR–Exposure gradient, the historical regime-associated difference, and the later post-ceiling-removal full-sample gradient for market funding relative to liabilities. Whiskers show CR2 95 percent confidence intervals.

The estimates carry substantial uncertainty: the regulated gradient has a CR2 95 percent interval of $[-0.330, 2.618]$, and the later gradient has an interval of $[-0.453, 0.404]$. The later point estimate is near zero, but this is not an equivalence result. The within-JSB standard deviation of exposure is 0.640 in upstream units, so the regulated coefficient corresponds to about 0.733 percentage points per within-JSB standard deviation. Table G.6 evaluates the same arithmetic for other exposure gaps and RRR differences; these are fitted-value comparisons, not structural counterfactuals.

4.2 Large banks and the cross-type comparison

The large-bank regime difference is 0.219, compared with -1.168 for joint-stock banks. Their difference is -1.388 , with conventional clustered $p = 0.009$, CR2 $p = 0.071$, and Rademacher and Webb p -values of 0.075 and 0.083. Thus small-cluster inference gives weaker precision for this cross-type comparison than the conventional benchmark. The comparison places the baseline pattern alongside the groups' different pre-reform liability structures; it does not identify ownership or a specific deposit-pricing mechanism.

5 RRR Support, Trends, and Timing

5.1 Policy support across periods

Figure 3 shows the downward shift in RRR support. Joint-stock banks' regulated and later ranges are 13.5–19.5 and 6.5–15.0 percent; the large-bank ranges are 15.5–21.5 and 9.5–17.0 percent. Their respective overlaps are only 13.5–15.0 and 15.5–17.0 percent. The JSB median falls from 15.5 to 9.5 percent and the large-bank median from 17.5 to 12.5 percent. The full-sample comparison therefore spans different regions of the reserve-policy distribution, allowing pricing conditions, nonlinearities, and other changes to contribute to the fitted gradient difference.

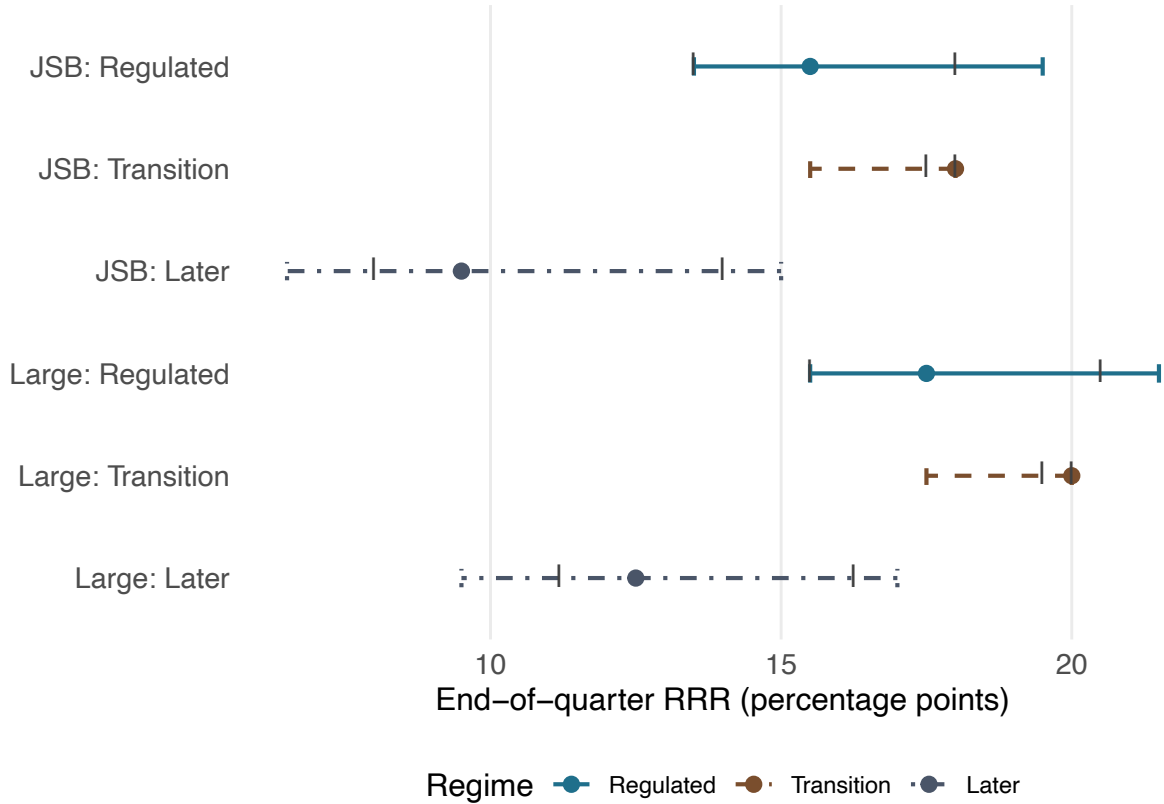


Figure 3: RRR support across historical regimes

Notes: Bars show minimum and maximum quarter-end RRR values, circles the medians, and ticks the 25th and 75th percentiles, by bank type and pricing regime. The within-type overlaps are [13.5,15.0] for joint-stock banks and [15.5,17.0] for large banks. The underlying series are monthly-path group benchmarks described in Section 3.3.

5.2 Support-restricted comparisons

Table 3 compares the baseline with three support diagnostics. Restricting each bank type to its overlapping RRR interval yields a JSB difference of -1.358 on 240 observations. Exact matching yields -1.374 on 217 observations. Both retain the baseline sign but are imprecise under CR2

inference. Exact overlap is particularly fragile: one quarter supplies the additional common RRR level, and removing it makes the target slope contrast non-estimable. Appendix E identifies that quarter and reports the contrast-specific estimability check.

The full-sample spline gives a difference of -0.799 when evaluated over the overlap, but it uses policy values outside that interval in estimation. These exercises describe sensitivity to policy support and functional form; they do not isolate deposit pricing from the accompanying policy environment.

Table 3: RRR-support diagnostics for the joint-stock exposure gradient

Diagnostic	JSB slope difference	CR2 p	Estimation obs.	Interpretation
Full historical specification	-1.168	0.066	880	Full historical support
Hard common-support linear comparison	-1.358	0.264	240	Type-specific within-type overlap
One-knot RRR spline, evaluated on overlap	-0.799	0.426	880	Full-sample spline, gradients averaged over overlap
Exact RRR matching	-1.374	0.227	217	Two values per type; fragile later support

Notes: All rows report the JSB gradient difference. Hard support uses RRR intervals $[13.5, 15.0]$ for JSBs and $[15.5, 17.0]$ for large banks. Exact matching uses shared RRR values within each type. The spline is estimated on all 880 observations with knot 15.0, then its RRR derivative is averaged on a 40-point overlap grid, using the left derivative at the knot; see Eq. (E.1). CR2 uses Satterthwaite degrees of freedom. The samples and support restrictions differ across rows.

5.3 Sensitivity to exposure-related trends

Table G.4 shows that a common exposure trend leaves the cross-type contrast similar, at -1.458 , but changes the JSB difference from -1.168 to 0.728 and raises the large-bank difference from 0.219 to 2.186 . Allowing trends to differ by bank type also reverses the cross-type point estimate, to 1.065 . The substantive regressors remain full rank, so this is specification sensitivity rather than exact collinearity. The baseline pattern cannot be read as a trend-robust estimate of the pricing reform’s effect.

5.4 Timing and pre-reform diagnostics

The cumulative endpoint exercise re-estimates the specification through successive fourth quarters. The cross-type contrast is positive at the 2015Q4, 2016Q4, and 2017Q4 endpoints (5.374 , 5.596 , and 4.968), becomes negative at 2018Q4 (-0.650), and reaches -1.388 by 2024Q4. The three early

estimates are distinguishable from zero conventionally, but not under CR2. The estimates around the sign change are also imprecise. Because each regression adds observations to the preceding sample, this grid does not date a structural break or trace an event-time response.

The leave-one-quarter-out exercise indicates that no individual quarter materially drives the full-sample contrast: its maximum absolute change is below 0.21. This does not establish robustness to removing multi-quarter episodes. Figure F.1 reports the endpoint estimates.

Pre-reform pseudo dates give mixed evidence. JSB slope differences at 2009Q2, 2010Q1, and 2010Q4 are 0.110, 0.344, and 0.123, all imprecise. The 2011Q2 pseudo date yields -1.740 with clustered $p = 0.143$, a larger negative point estimate than the full-sample difference. Figure F.2 retains all four comparisons. The flexible-gradient paths in Appendix F provide related functional-form diagnostics.

6 Liability-Management Evidence

6.1 Customer deposits and liability composition

Table 4 compares liability outcomes on the same 880 bank-quarters. The negative JSB regime difference in bonds-plus-repo funding is not accompanied by a precisely estimated increase in customer-deposit funding: the customer-deposit difference is 0.012. The broader noncustomer measure is also imprecise, at -0.215 . These results document the baseline pattern for the specified market-funding outcome but do not establish one-for-one substitution into deposits or a tested difference between outcomes.

Customer deposits, noncustomer funding, and residual liabilities form an accounting partition and sum to 100 percent. Market funding, reported interbank liabilities, and repos overlap and cannot be summed as additional components. Appendix Table H.1 reports the separate outcome-specific samples; those estimates are not an accounting decomposition of the common-sample coefficient.

Table 4: Liability composition across regimes

Outcome	JSB difference	Cluster p	CR2 p	Cross-type	CR2 p
<i>Panel A. Accounting partition (shares sum to 100 by construction)</i>					
Customer deposits	0.012	0.985	0.986	0.977	0.396
Noncustomer funding (dedup)	-0.215	0.519	0.572	-1.922	0.323
Residual liabilities	0.203	0.614	0.659	0.946	0.410
<i>Panel B. Overlapping diagnostic outcomes (do not sum)</i>					
Market funding (bonds+repo)	-1.168	0.019	0.066	-1.388	0.071
Interbank liabilities	0.165	0.513	0.565	-1.255	0.394
Repo liabilities	-0.788	0.084	0.153	-0.720	0.274

Notes: All rows are estimated on the same 880 baseline observations. Panel A is an accounting partition: noncustomer funding is the union of bonds payable and reported interbank liabilities with repo counted once, and residual liabilities are defined as total liabilities less customer deposits and noncustomer funding, so the three shares sum to 100 by construction. Panel B reports outcomes that overlap one another and with Panel A: market funding is bonds payable plus repo, reported interbank liabilities already include repo, and repo is shown separately. Panel B rows therefore should not be summed or read as a decomposition. Under conventional bank-clustered inference at the 5 percent level, only the market-funding row is distinguishable from zero; its CR2 p -value is 0.066.

Annual disclosures provide further descriptive evidence. On the common 156 bank-year sample, the cross-type market-funding contrast is -1.093 (CR2 $p = 0.429$), and the reported deposit-rate composite contrast is -0.055 (CR2 $p = 0.462$). The annual market-funding range under individual-bank exclusions crosses zero. Appendix J reports the rate components and annual design. This evidence does not isolate a deposit-pricing channel.

6.2 Market-funding definitions and NCDs

The headline measure combines bonds payable and repo liabilities, two separately observed non-deposit funding categories. The broader measure adds interbank deposits and borrowing without counting repos twice. Legal NCD access and first observed issuance leave the matched-sample contrast negative. Realized usage is a post-adoption choice observed on a shorter sample. Appendix I reports the precise augmentations and paired baseline estimates, keeping sample composition separate from the addition of NCD terms. Neither these comparisons nor the liability outcomes identify a unique substitution mechanism.

7 Additional Robustness and Inference

7.1 Alternative measurement and bank influence

All alternative exposure definitions end by 2011Q4. Changing the averaging window or using the last pre-reform report gives cross-type contrasts from -1.573 to -1.306 in the baseline scale. Within-type standardization and the two residualized measures also give negative contrasts, but change exposure units, and the latter additionally exclude Zheshang because its pre-reform log-assets measure is unavailable. Their raw coefficients are not common-unit magnitude comparisons. Table G.2 states each transformation.

Changing the regime index and RRR timing convention gives contrasts from -1.425 to -1.354 . Individual-bank exclusions give -1.650 to -0.959 . Table 5 summarizes these results together with the adverse support and trend diagnostics. They describe the stability of the baseline pattern under specific checks, not uniform robustness to every specification.

Table 5: Identification and sensitivity summary

Diagnostic	Estimate or range	CR2 p	Interpretation
Full-sample JSB slope difference	-1.168	0.066	Historical full-sample pattern
Later JSB full-sample slope	-0.024	0.888	Near-zero later gradient
JSB minus large-bank contrast	-1.388	0.071	Funding-model comparison
Alternative exposure windows	[-1.573, -1.306]	—	Fixed baseline scale; all pre-2012
Alternative indices and RRR timing	[-1.425, -1.354]	—	Policy-variable construction
Leave-one-bank-out contrasts	[-1.650, -0.959]	—	Bank influence
Within-type exposure permutation	$p = 0.012$	—	Sensitivity diagnostic, not randomization inference
Hard common support	-1.358	0.264	Type-specific overlap, 240 obs.
Exact RRR overlap	-1.374	0.227	Two shared values; thin later support
Common exposure trend	-1.458	0.070	Contrast held; JSB within-type difference reverses
Type-specific exposure trend	1.065	0.342	Reverses the cross-type sign; see Table G.4
2018Q4 cumulative endpoint	-0.650	0.522	Endpoint at which the sign changes
Noncustomer funding, each component once	-0.215	0.572	Funding definition, canonical sample

Notes: Ranges report the minimum and maximum within each stated family and have no single p -value. Alternative windows include 2009–2011, 2010–2011, 2011, and the last pre-reform report. Within-type standardized and residualized exposures use different units and are reported separately in Table G.2. The permutation row reports a two-sided tail proportion from 999 within-type exposure reassignments; exposure was not randomly assigned.

7.2 Inference

The cross-type contrast’s conventional p -value is 0.009, compared with 0.071 under CR2 and 0.075 and 0.083 under the two wild-cluster procedures. The JSB difference has CR2 $p = 0.066$ but smaller bootstrap p -values, illustrating that the methods do not have a universal ordering. These procedures quantify uncertainty around the same fitted coefficients, rather than provide independent estimates confirming a sign.

A within-type exposure permutation yields a two-sided tail proportion of 0.012 from 999 re-assignments. Exposure is not randomly assigned, so this is a permutation sensitivity diagnostic, not a design-based randomization test. Its interpretation does not remove the policy-support and exposure-trend limitations. The paper reports magnitudes and method-specific uncertainty together rather than relying on a single significance threshold.

8 Conclusion

This paper compares exposure-based bank-funding gradients across deposit-pricing regimes in a quarterly panel of fourteen Chinese banks over 2008–2024. In the baseline levels specification, the joint-stock-bank market-funding gradient is 1.144 in the regulated period and -0.024 in the later period. The regime difference is -1.168 , and the joint-stock-minus-large-bank contrast is -1.388 . These magnitudes describe how the fitted funding gap between banks one upstream standard deviation apart in pre-reform time-deposit exposure changes with a one-percentage-point difference in the RRR. They are conditional historical associations.

The broader implication concerns the interpretation of exposure-based evidence. A funding characteristic measured before a regulatory transition can remain predetermined while its fitted relationship with bank funding varies across the surrounding policy environment. Comparing estimates across periods therefore requires attention to the region of policy support and the evolution of exposure-related bank characteristics. The baseline comparison provides a concrete example of why the stability assumption matters. The funding-model comparison and illustrative liability-choice framework help organize this question, without delivering a structural estimate that can be transferred unchanged to another banking system.

The evidence does not distinguish the contribution of deposit-price liberalization from concurrent changes in RRR support and exposure-related trends. Small-cluster uncertainty is substantial, and the deposit-side and NCD evidence leaves the funding mechanism unresolved. Product-level pricing data and policy episodes with richer common support would help separate these explanations. Those extensions are outside the present historical comparison.

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Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The analysis combines bank financial statements obtained through Tonghuashun iFinD, supplementary East Money and Sina Finance displays, and public bank and policy disclosures. Access to vendor extracts is subject to the respective providers' terms. Construction code and source documentation are available from the corresponding author; redistribution of vendor data depends on the applicable permissions.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used OpenAI ChatGPT and Codex to assist with language editing, literature organization, code development and review, data verification, data-derived visualization, manuscript structuring, and submission checks. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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A Institutional Chronology and Regime-Index Construction

The primary pure deposit-ceiling index maps six direct deposit-ceiling or deposit-product-scope changes into a day-weighted quarterly series. Let direct deposit-pricing events $j = 1, \dots, 6$ have effective dates d_j . The equal-step daily index is

$$\ell_d^{equal} = \frac{1}{6} \sum_{j=1}^6 \mathbf{1}\{d \geq d_j\}, \quad (\text{A.1})$$

and for quarter t with calendar-day set D_t the regression index is

$$L_t^{equal} = \frac{1}{|D_t|} \sum_{d \in D_t} \ell_d^{equal}. \quad (\text{A.2})$$

For the secondary index, let cumulative levels be $c = (0.2, 0.4, 0.6, 0.8, 0.9, 1.0)$ and increments $w = (0.2, 0.2, 0.2, 0.2, 0.1, 0.1)$. Then

$$\ell_d^{cap} = \sum_{j=1}^6 w_j \mathbf{1}\{d \geq d_j\}, \quad L_t^{cap} = |D_t|^{-1} \sum_{d \in D_t} \ell_d^{cap}. \quad (\text{A.3})$$

This alternative coding gives each ceiling-multiplier change weight 0.2 and each of the two final scope removals weight 0.1. It is a transparent sensitivity index, not a cardinal measure of economic liberalization. The effective day is included. Table A.1 gives the cumulative daily values; quarterly values are calendar-day averages. The June 2015 large-denomination certificate-of-deposit rules do not enter either pure deposit-ceiling index.

Table A.1: Direct deposit-ceiling reforms and cumulative daily index levels

Effective date	Rule change	Equal-step	Cap/scope
2012-06-08	Ceiling rises from 1 to 1.1 times benchmark	1/6	0.2
2014-11-22	Ceiling rises to 1.2 times benchmark	2/6	0.4
2015-03-01	Ceiling rises to 1.3 times benchmark	3/6	0.6
2015-05-11	Ceiling rises to 1.5 times benchmark	4/6	0.8
2015-08-26	Ceiling removed for time deposits longer than one year	5/6	0.9
2015-10-24	Remaining RMB deposit-rate ceilings removed	6/6	1.0

Notes: Entries are cumulative daily levels from each effective date, not quarterly averages. Quarterly indices average all calendar days; 2016Q1 is the first full quarter at one. Sources: PBoC, *Highlights of China's Monetary Policy in the Second Quarter of 2012*; *Annual Report 2014*; the 2015 monetary-policy chronology; and the 2015Q4 monetary-policy report (People's Bank of China, 2012, 2014, 2015, 2016). Large-denomination customer-CD rules of June 2015 are excluded.

Table A.2: Institutional chronology of concurrent reforms

Reform family	Timing and incidence	Available measure	Role in the analysis
Deposit-rate liberalization	National sequence of ceiling and product-scope reforms, 2012Q2–2015Q4; the index reaches one in 2016Q1, the first full quarter after formal completion.	Quarter-level liberalization index.	Main policy variable; bank-type-by-quarter fixed effects absorb common level effects.
NCD framework	Bank-level legal access, adoption, and realized use from 2013Q4 onward.	Legal access date, first issuance, lagged balances or usage, and NCD quantities.	Matched-sample competing-explanation and mechanism checks.
Large-denomination certificates of deposit	Common national product reform in 2015Q2 with heterogeneous take-up.	Policy date and component outcomes.	Concurrent customer-deposit product reform kept outside the pure deposit-ceiling index.
Deposit insurance	Common national implementation in 2015Q2.	Policy date.	Fixed effects absorb common level components; exposure-heterogeneous incidence is not removed.
Liquidity and macroprudential reforms	National and bank-level regulatory changes after 2015.	Policy chronology; no separate quantitative reform measure.	Common level components enter fixed effects; heterogeneous effects remain a limitation.
Targeted RRR and later deleveraging	Sample-period changes in reserve policy and balance-sheet regulation.	Quarter-end, quarter-start, and average RRR paths.	Group RRR benchmarks enter directly; bank-specific concessions and exposure-heterogeneous reforms are not separately identified.

B Data, Sample, Exposure, and Descriptive Statistics

Appendix Table B.1 lists the fourteen nationwide banks and sample coverage. Appendix Table B.2 reports full-sample funding composition, and Appendix Table B.3 documents pre-reform exposure correlations. Figure B.1 displays the predetermined exposure measure by bank.

Table B.1: Bank sample and predetermined exposure

Bank	Type	Exposure	Panel first	MF first
Agricultural Bank of China	Large	-0.150	2008Q4	2008Q4
Bank of China	Large	0.549	2008Q1	2008Q2
Bank of Communications	Large	0.523	2008Q1	2008Q1
China Construction Bank	Large	0.190	2008Q1	2008Q1
Industrial and Commercial Bank of China	Large	0.303	2008Q1	2008Q1
China Everbright Bank	JSB	1.002	2008Q4	2008Q4
China CITIC Bank	JSB	0.912	2008Q1	2008Q1
China Merchants Bank	JSB	0.275	2008Q1	2008Q1
China Minsheng Banking Corp.	JSB	1.095	2008Q1	2008Q4
China Zheshang Bank	JSB	1.603	2008Q2	2012Q4
Hua Xia Bank	JSB	-0.191	2008Q1	2008Q1
Industrial Bank	JSB	0.092	2008Q1	2008Q1
Ping An Bank	JSB	-0.262	2008Q1	2008Q1
Shanghai Pudong Development Bank	JSB	0.331	2008Q1	2008Q1

Notes: Large denotes the five large state-owned banks; JSB denotes the nine national joint-stock banks. Exposure uses the fixed upstream scale in Table B.4. Panel first is the earliest bank-quarter row; MF first is the first observed market-funding outcome. Both series end in 2024Q4 for every bank.

Table B.2: Funding composition by deposit-pricing regime

Type	Regime	Share	Mean	Asset-weighted	SD	Obs.	Banks
Large state-owned	Pre	Deposits / liabilities	83.46	84.71	5.46	73	5
Large state-owned	Transition	Deposits / liabilities	80.52	81.54	5.25	75	5
Large state-owned	Full	Deposits / liabilities	77.32	78.93	6.42	180	5
Large state-owned	Pre	Noncustomer funding / liabilities	13.31	11.97	5.07	73	5
Large state-owned	Transition	Noncustomer funding / liabilities	14.40	13.14	5.22	75	5
Large state-owned	Full	Noncustomer funding / liabilities	15.03	14.52	3.67	180	5
Large state-owned	Pre	Residual liabilities / liabilities	3.23	3.32	1.60	73	5
Large state-owned	Transition	Residual liabilities / liabilities	5.07	5.32	1.41	75	5
Large state-owned	Full	Residual liabilities / liabilities	7.65	6.55	3.49	180	5
National joint-stock	Pre	Deposits / liabilities	77.60	77.83	7.58	120	8
National joint-stock	Transition	Deposits / liabilities	69.89	69.47	7.21	121	9
National joint-stock	Full	Deposits / liabilities	63.56	63.86	6.96	311	9
National joint-stock	Pre	Noncustomer funding / liabilities	19.97	19.71	7.68	120	8
National joint-stock	Transition	Noncustomer funding / liabilities	27.03	27.31	7.13	121	9
National joint-stock	Full	Noncustomer funding / liabilities	30.31	30.11	7.23	311	9
National joint-stock	Pre	Residual liabilities / liabilities	2.43	2.46	1.33	120	8
National joint-stock	Transition	Residual liabilities / liabilities	3.08	3.22	1.24	121	9
National joint-stock	Full	Residual liabilities / liabilities	6.14	6.03	1.90	311	9

Notes: All entries use the canonical 880-observation estimation sample, not each outcome's own available sample. Regimes use the primary equal-step index: pre means $L = 0$, transition $0 < L < 1$, and full $L = 1$. The three shares form an accounting partition and sum to 100 within each cell by construction; noncustomer funding is bonds payable plus reported interbank liabilities, which already include repo liabilities.

Table B.3: Predetermined exposure: distribution and correlations

Measure or group	Banks	Mean	SD	Min	Median	Max	Corr. with baseline
<i>A. Distribution of the baseline exposure</i>							
All banks	14	0.448	0.542	-0.262	0.317	1.603	
National joint-stock	9	0.540	0.640	-0.262	0.331	1.603	
Large state-owned	5	0.283	0.285	-0.150	0.303	0.549	
<i>B. Correlations with pre-reform characteristics</i>							
Pre-period deposits / liabilities							0.009
Pre-period log assets							-0.020
<i>C. Alternative predetermined exposure correlations</i>							
Exposure window 2009Q1–2011Q4							0.980
Exposure window 2010Q1–2011Q4							0.958
Exposure window 2011Q1–2011Q4							0.950
Last pre-reform observation							0.961
Within-type standardized							0.925
Size-residualized							1.000
Size- and type-residualized							0.955

Notes: Panel A describes the single baseline exposure object used throughout the paper. Panel B reports correlations with bank-level pre-2012 means of the stated characteristic. Panel C correlates the baseline with the alternative predetermined constructions of Table G.2, all of which use only information dated 2011Q4 or earlier.

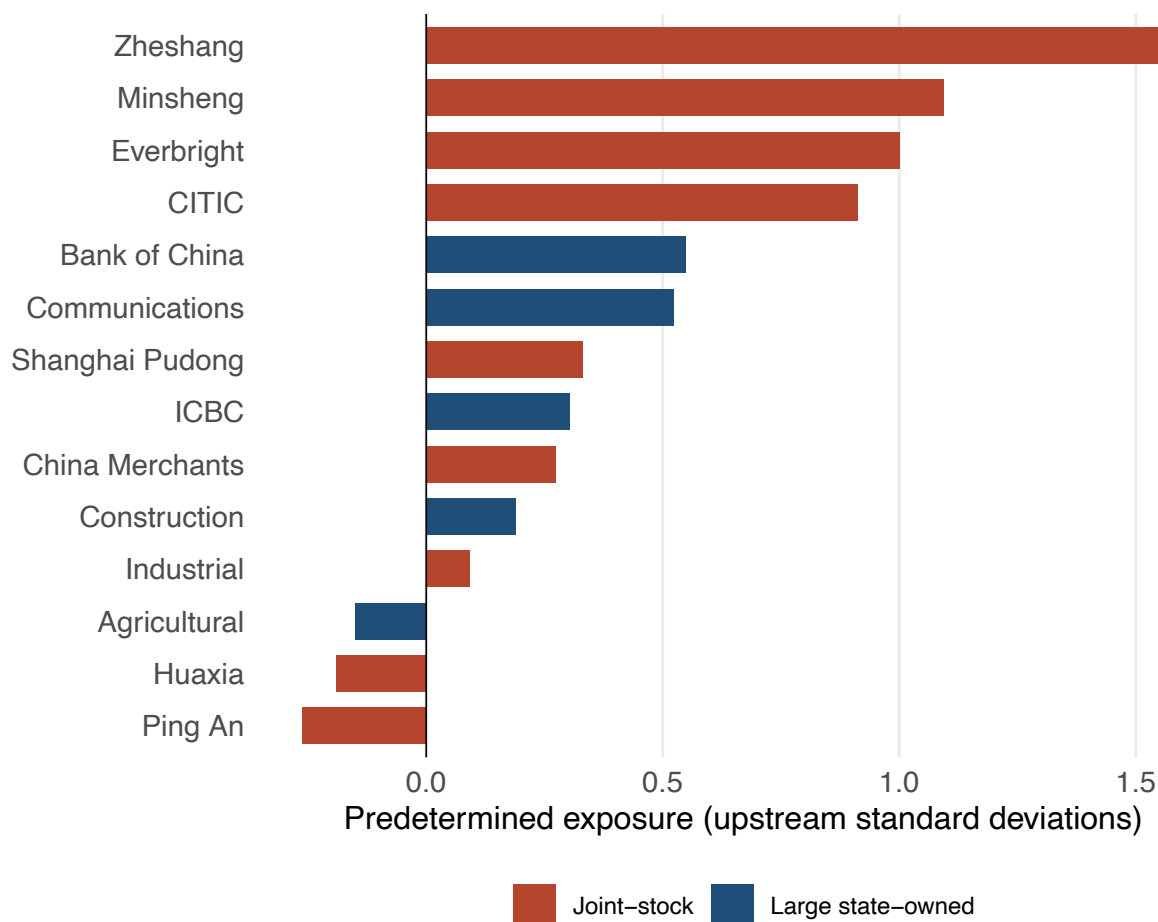


Figure B.1: Predetermined time-deposit exposure by bank

Notes: Bars report the standardized predetermined exposure used in the baseline specification, defined as each bank's 2008Q1–2011Q4 mean of (corporate time deposits + personal time deposits) divided by customer deposits, standardized with the upstream pre-reform roster mean 0.4241 and standard deviation 0.1217. Appendix Table B.4 reports the raw share, the number of contributing quarters and the standardized value for every bank, and reproduces each value shown here.

Table B.4: Exposure construction, bank by bank

Bank	Type	Pre-reform quarters	Raw time-deposit share	Standardized exposure
CZBank	Joint-stock	7	61.91	1.603
CMBC	Joint-stock	7	55.73	1.095
CEB	Joint-stock	8	54.60	1.002
CITIC	Joint-stock	8	53.50	0.912
BOC	Large	8	49.09	0.549
BOCOM	Large	16	48.78	0.523
SPDB	Joint-stock	13	46.44	0.331
ICBC	Large	8	46.10	0.303
CMB	Joint-stock	9	45.75	0.275
CCB	Large	8	44.72	0.190
IB	Joint-stock	7	43.53	0.092
ABC	Large	6	40.58	-0.150
HXB	Joint-stock	7	40.09	-0.191
PAB	Joint-stock	9	39.22	-0.262

Notes: The raw measure is the bank mean over 2008Q1–2011Q4 of (corporate time deposits + personal time deposits) divided by customer deposits, expressed in percent, computed over the quarters in which that ratio is observed. Time-deposit components are disclosed mainly at half-year and year-end, which is why the number of contributing quarters ranges from six to sixteen. The standardized measure is $(raw - \mu)/\sigma$ with $\mu = 0.4241$ and $\sigma = 0.1217$ on the fraction scale (42.41 percent and 12.17 percentage points), taken from the upstream pre-reform standardization roster rather than from the fourteen estimation banks, whose own mean and standard deviation are 47.86 percent and 6.59 percentage points. Every baseline standardized exposure value shown elsewhere in the paper is reproduced by this rule.

B.1 Exposure scale and inference

The upstream standardization uses an equally weighted mean and sample standard deviation across 41 finite bank-level exposure values:

$$\mu_0 = 0.424099679174176, \quad \sigma_0 = 0.121657116159610. \quad (\text{B.1})$$

These constants are held fixed when the fourteen banks' raw exposure values are updated after source reconciliation. The upstream bank identifiers are 000001.SZ, 001227.SZ, 002142.SZ, 002807.SZ, 002839.SZ, 002936.SZ, 002948.SZ, 002958.SZ, 002966.SZ, 600000.SH, 600015.SH, 600016.SH, 600036.SH, 600908.SH, 600919.SH, 600926.SH, 600928.SH, 601009.SH, 601077.SH, 601128.SH, 601166.SH, 601169.SH, 601187.SH, 601229.SH, 601288.SH, 601328.SH, 601398.SH, 601528.SH, 601577.SH, 601665.SH, 601818.SH, 601825.SH, 601838.SH, 601860.SH, 601916.SH, 601939.SH, 601963.SH, 601988.SH, 601997.SH, 601998.SH, 603323.SH. The roster covers banks with available pre-2012 time-deposit disclosures, including banks outside the fourteen-bank

regression sample. Its complete values and inclusion periods are preserved in the replication files listed in the variable dictionary.

For wild-cluster inference, the null contrast is imposed in the original linear model. Unadjusted restricted residuals are multiplied by one weight per bank, the unrestricted model is refitted, and the contrast is studentized with its conventional cluster-robust standard error. Single-coefficient reparameterizations of sums or differences preserve the same fitted model. Rademacher weights enumerate all 16,384 pooled bank sign patterns. A within-type coefficient effectively uses only its own block: five large-bank signs give 32 patterns and nine JSB signs give 512. Two-sided tests include ties within $10^{-8} \max(1, |t|)$. Webb weights use 9,999 draws, with both random-number generators seeded at 20260812. No invalid draws are discarded. This finite-support calculation does not imply exact finite-sample test size under arbitrary errors.

Table B.5: Customer-deposit exposure-gradient decomposition

Estimand	Estimate (SE)	<i>p</i> : CR1	<i>p</i> : CR2	<i>p</i> : WCR-R	<i>p</i> : WCR-W
<i>Panel A. Customer deposits / total assets</i>					
JSB regulated gradient	-0.510 (0.732)	0.498	0.554	0.605	0.587
JSB regime difference	-0.113 (0.633)	0.861	0.877	0.902	0.903
JSB later gradient	-0.623 (0.336)	0.087	0.132	0.207	0.195
Large-bank regime difference	-1.267 (0.482)	0.021	0.128	0.0625	0.117
JSB minus large-bank difference	1.154 (0.796)	0.171	0.282	0.294	0.306
<i>Panel B. Customer deposits / total liabilities</i>					
JSB regulated gradient	-0.648 (0.762)	0.410	0.475	0.500	0.494
JSB regime difference	-0.004 (0.652)	0.995	0.996	1.000	0.996
JSB later gradient	-0.652 (0.369)	0.101	0.146	0.199	0.203
Large-bank regime difference	-1.082 (0.631)	0.110	0.245	0.375	0.349
JSB minus large-bank difference	1.078 (0.907)	0.256	0.360	0.365	0.366

Notes: The panels use different denominators and are separate regressions. Panel A uses 892 observations and Panel B 892, each with nine joint-stock and five large banks. Outcomes are percentage shares; exposure and RRR units match Table 2. All rows include bank and bank-type-by-quarter fixed effects. Standard errors are conventional bank-clustered; the four tests follow Table 2. Only five banks directly inform the large-bank difference. For the deposits/assets difference, the Rademacher tail contains the two tied extreme patterns out of 32 effective signs, giving $p = 0.0625$.

C Sample Coverage and Source Reconciliation

The fourteen-bank grid contains 952 potential bank-quarters. Source records exist for 922, and 880 have the market-funding outcome. Thirty absent records belong to Agricultural Bank, Everbright, and Zheshang before 2016Q2. Agricultural Bank and Everbright have year-end observations in 2008–2009 and continuous observations from 2010Q2; Zheshang has primarily half-year and year-end observations before continuous coverage begins in 2016Q2. The pattern is consistent with changing disclosure frequency. Listing dates alone do not establish the cause of each missing report.

The source financial statements were accessed through Tonghuashun iFinD, with supplementary consolidated-statement values from East Money and corroborating displays from Sina Finance. A trading-day request grid in the original extract omitted 21 non-trading-day quarter-ends for several liability fields. Re-extraction by statement period end supplied affected components for 287 bank-quarter rows, of which 275 meet all outcome requirements. These observations expand the market-funding sample from 605 to 880 and restore coverage of all 68 calendar quarters. The final panel retains every available valid outcome rather than interpolating missing financial statements.

The iFinD fields for bonds payable, repos, and interbank liabilities map respectively to East Money’s BOND_PAYABLE, SELL_REPO_FINASSET, and the sum of IOFI_DEPOSIT, BORROW_FUND, and SELL_REPO_FINASSET. Both extracts use renminbi at statement scale. The interbank mapping agrees in 905 of 909 overlapping cells. Where overlapping displays disagreed materially, comparison with Sina corroborated 23 value corrections. Agreement between displays does not establish independence of their upstream sources. The replication audit preserves cell-level comparisons and the separate data and exposure updates.

REPORT_DATE is the statement period end, not a release or restatement timestamp. The retained East Money extract contains no duplicate bank-quarter records. The extraction routine would keep the first returned record if duplicates occurred, without a chronological restatement rule. Stable row identifiers and file hashes freeze the observations used; they do not reconstruct historical publication vintages.

Forty-two existing rows still lack the outcome: thirty lack total liabilities, thirteen lack repos, and five lack bonds payable, with overlapping reasons. These gaps are concentrated in Zheshang (21 rows) and Everbright (8 rows). The policy variables, exposure, and fixed effects introduce no additional baseline exclusions. Exposure uses its own pre-2012 reporting sample, with six to sixteen reports per bank, mainly at half-year and year-end. Table C.1 reports coverage by bank and year.

Table C.1: Bank-by-quarter coverage of the market-funding outcome, repaired sample

Bank	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	Total
ABC	1	1	2	•	•	•	•	•	•	•	•	•	•	•	•	•	•	60
BOC	3	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	67
BOCOM	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	68
CCB	•	3	3	•	•	•	•	•	•	•	•	•	•	•	•	•	•	66
ICBC	3	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	67
CEB	1	1	1	1	2	3	•	•	•	•	•	•	•	•	•	•	•	53
CITIC	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	68
CMB	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	68
CMBC	1	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	65
CZBank	–	–	–	–	1	1	1	2	2	2	1	2	•	3	•	•	•	31
HXB	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	68
IB	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	68
PAB	•	•	•	•	•	•	•	2	3	•	•	•	•	3	3	•	•	63
SPDB	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	•	68
Total	41	45	46	49	51	52	53	52	53	54	53	54	56	54	55	56	56	880

Notes: Each cell reports the number of quarters in that calendar year for which the bank has a nonmissing market-funding ratio. • denotes all four quarters; – denotes none. The outcome requires bonds payable, repo liabilities and total liabilities. All 68 calendar quarters are represented. The 30 grid cells with no bank-quarter row at all (Agricultural Bank and Everbright before 2010Q2, Zheshang before 2016Q2) are counted as zero here.

D Liability-Choice Framework

The following derivation formalizes the comparative-static role of a pricing constraint. It is not structurally estimated, and it does not map the historical regime coefficient into a deposit-pricing parameter.

D.1 Derivation

The framework summarized in Section 2.4 is a local funding-composition model. It is not used for structural estimation. A bank of type $j \in \{G, J\}$ meets a fixed gross local funding target F (with other balance-sheet items held fixed) with customer deposits d and market funding w :

$$d + w = F, \quad 0 \leq d \leq F, \quad w = F - d. \quad (\text{D.1})$$

The deposit cost and market-funding cost are

$$C_j^D(d; \alpha, L, \theta) = [\mu_j + \alpha R_j(L, \theta)] d + \frac{K_j(L, \theta)}{2} d^2, \quad (\text{D.2})$$

$$C_j^W(w) = \xi_j w + \frac{\nu_j}{2} w^2, \quad (\text{D.3})$$

where $\alpha \in [0, 1]$ is the reserve rate, $L \in [0, 1]$ indexes pricing flexibility, and $\theta \in [0, 1]$ is a raw time-deposit share, distinct from standardized empirical exposure E . The functions used in the text are

$$R_j(L, \theta) = \rho_{0j} + \rho_{1j}\theta(1 - L), \quad K_j(L, \theta) = \kappa_j \exp(-\lambda_j L\theta), \quad (\text{D.4})$$

with $\kappa_j, \nu_j > 0$, $\rho_{0j} \geq 0$, $\rho_{1j} > 0$, and $\lambda_j \geq 0$. These conditions imply $R_j \geq 0$ and positive cost curvature. The interior comparative statics require $0 < \xi_j + \nu_j F - \mu_j - \alpha R_j < F(K_j + \nu_j)$.

Substituting $w = F - d$, the bank minimizes

$$\Phi_j(d) = [\mu_j + \alpha R_j(L, \theta)]d + \frac{K_j(L, \theta)}{2}d^2 + \xi_j(F - d) + \frac{\nu_j}{2}(F - d)^2. \quad (\text{D.5})$$

For an interior solution, the first-order condition is

$$\frac{\partial \Phi_j}{\partial d} = \mu_j + \alpha R_j(L, \theta) + K_j(L, \theta)d - \xi_j - \nu_j(F - d) = 0. \quad (\text{D.6})$$

The second derivative is

$$\frac{\partial^2 \Phi_j}{\partial d^2} = K_j(L, \theta) + \nu_j > 0, \quad (\text{D.7})$$

so the interior problem is strictly convex and the first-order condition is sufficient.

Let $D_j(L, \theta) = K_j(L, \theta) + \nu_j$. The unconstrained interior solution is

$$d_j^*(\alpha, L, \theta) = \frac{\xi_j + \nu_j F - \mu_j - \alpha R_j(L, \theta)}{D_j(L, \theta)}, \quad (\text{D.8})$$

$$w_j^*(\alpha, L, \theta) = F - d_j^*(\alpha, L, \theta) = \frac{K_j(L, \theta)F - \xi_j + \mu_j + \alpha R_j(L, \theta)}{D_j(L, \theta)}. \quad (\text{D.9})$$

Because R_j and K_j do not depend on α ,

$$\frac{\partial w_j^*}{\partial \alpha} = \frac{R_j(L, \theta)}{D_j(L, \theta)}. \quad (\text{D.10})$$

This is a funding-level derivative. Section 2.4 supplies the conversion to empirical share and policy units.

The exposure gradient in that RRR slope is

$$\frac{\partial^2 w_j^*}{\partial \alpha \partial \theta} = \frac{R_{j,\theta}(L, \theta)D_j(L, \theta) - R_j(L, \theta)K_{j,\theta}(L, \theta)}{D_j(L, \theta)^2}. \quad (\text{D.11})$$

Using

$$R_{j,\theta}(L, \theta) = \rho_{1j}(1 - L), \quad K_{j,\theta}(L, \theta) = -\lambda_j L K_j(L, \theta), \quad (\text{D.12})$$

gives

$$\frac{\partial^2 w_j^*}{\partial \alpha \partial \theta} = \frac{\rho_{1j}(1 - L) [K_j(L, \theta) + \nu_j] + \lambda_j L K_j(L, \theta) R_j(L, \theta)}{[K_j(L, \theta) + \nu_j]^2}. \quad (\text{D.13})$$

In the regulated-pricing environment, $L = 0$, so $K_j(0, \theta) = \kappa_j$ and

$$\left. \frac{\partial^2 w_j^*}{\partial \alpha \partial \theta} \right|_{L=0} = \frac{\rho_{1j}}{\kappa_j + \nu_j}. \quad (\text{D.14})$$

In the fully flexible pricing environment, $L = 1$, so $R_j(1, \theta) = \rho_{0j}$ and $K_j(1, \theta) = \kappa_j \exp(-\lambda_j \theta)$.

The exposure gradient becomes

$$\left. \frac{\partial^2 w_j^*}{\partial \alpha \partial \theta} \right|_{L=1} = \frac{\lambda_j \kappa_j \exp(-\lambda_j \theta) \rho_{0j}}{[\kappa_j \exp(-\lambda_j \theta) + \nu_j]^2}. \quad (\text{D.15})$$

The model's market-funding exposure gradient is smaller in the flexible-pricing environment when

$$\frac{\lambda_j \kappa_j \exp(-\lambda_j \theta) \rho_{0j}}{[\kappa_j \exp(-\lambda_j \theta) + \nu_j]^2} < \frac{\rho_{1j}}{\kappa_j + \nu_j}. \quad (\text{D.16})$$

A special case is $\lambda_j = 0$ or $\rho_{0j} = 0$: the flexible-pricing gradient is zero and is strictly below the regulated gradient under the maintained $\rho_{1j} > 0$. More generally, attenuation requires the remaining flexibility channel on the left-hand side to be smaller than the regulated-pricing exposure-specific reserve wedge on the right-hand side.

Because $d_j^* = F - w_j^*$, the corresponding deposit response has the opposite sign:

$$\frac{\partial^2 d_j^*}{\partial \alpha \partial \theta} = -\frac{\partial^2 w_j^*}{\partial \alpha \partial \theta}. \quad (\text{D.17})$$

This opposite-sign relation holds locally under the fixed gross funding target. It is not an accounting restriction on empirical outcomes with additional liability categories or different samples.

The empirical design does not separately identify quoted-rate and product-composition effects. The deposit-cost schedule represents a possible deposit-side adjustment margin; the annual results do not identify it.

The constrained solution is the projection of the unconstrained solution onto the feasible interval:

$$d_j^c = \min \{F, \max \{0, d_j^*\}\}, \quad w_j^c = F - d_j^c. \quad (\text{D.18})$$

Equivalently, the lower-deposit corner $d = 0$ applies when

$$\mu_j + \alpha R_j(L, \theta) - \xi_j - \nu_j F \geq 0, \quad (\text{D.19})$$

and the full-deposit corner $d = F$ applies when

$$\mu_j + \alpha R_j(L, \theta) + K_j(L, \theta)F - \xi_j \leq 0. \quad (\text{D.20})$$

Within a corner, the local derivative of w_j^c with respect to α is zero until a parameter change moves the bank back into the interior. The empirical regressions should therefore be read as reduced-form local slope estimates, not as direct estimates of the structural primitives.

The model is deliberately partial equilibrium. It holds F , assets, equity, other liabilities, market prices, and aggregate quantities fixed. Its role is to show why a pricing constraint can make an exposure gradient in market funding depend on the institutional environment. It does not establish that the historical change in the empirical gradient was generated by deposit pricing, nor does it identify welfare, aggregate monetary transmission, or counterfactual bank balance sheets.

E RRR Support and Nonlinear Diagnostics

The within-type overlap is 13.5–15.0 percent for JSBs and 15.5–17.0 percent for large banks. The hard-support comparison includes 240 observations. Exact matching retains the shared values 13.5

and 15.0 for JSBs and 15.5 and 17.0 for large banks, giving 217 observations. The full exact sample permits estimation of the target slope contrast. The later-regime observations at 15.0 (JSB) and 17.0 (large) occur in 2018Q3. Deleting that quarter makes the contrast non-estimable, as checked by projecting its vector onto the design-matrix row space. Redundant fixed-effect columns alone are not used to declare a substantive contrast non-estimable.

The spline uses R and $(R - \kappa)_+ = \max(R - \kappa, 0)$, each interacted with exposure, the regime index, and bank type, with the same fixed effects. The knot $\kappa = 15.0$ is the pooled median RRR. All 880 observations enter estimation. For each type, define

$$\begin{aligned} h_g(R, L) &= \frac{\partial \hat{Y}}{\partial E} = b_{1g}R + b_{2g}L + b_{3g}RL + c_{1g}(R - \kappa)_+ + c_{3g}L(R - \kappa)_+, \\ k_g(R, L) &= \frac{\partial h_g(R, L)}{\partial R} = b_{1g} + b_{3g}L + \{c_{1g} + c_{3g}L\}\mathbf{1}\{R > \kappa\}. \end{aligned} \quad (\text{E.1})$$

Table 3 reports the average of $k_J(R, 1) - k_J(R, 0)$ on 40 equally spaced points in the JSB overlap. At the knot the left derivative is used. The derivative is piecewise constant: the JSB grid is at or below the knot and the large-bank grid is above it. Nevertheless, observations outside the overlap can affect the fitted coefficients. This is a full-sample functional-form diagnostic evaluated on overlap.

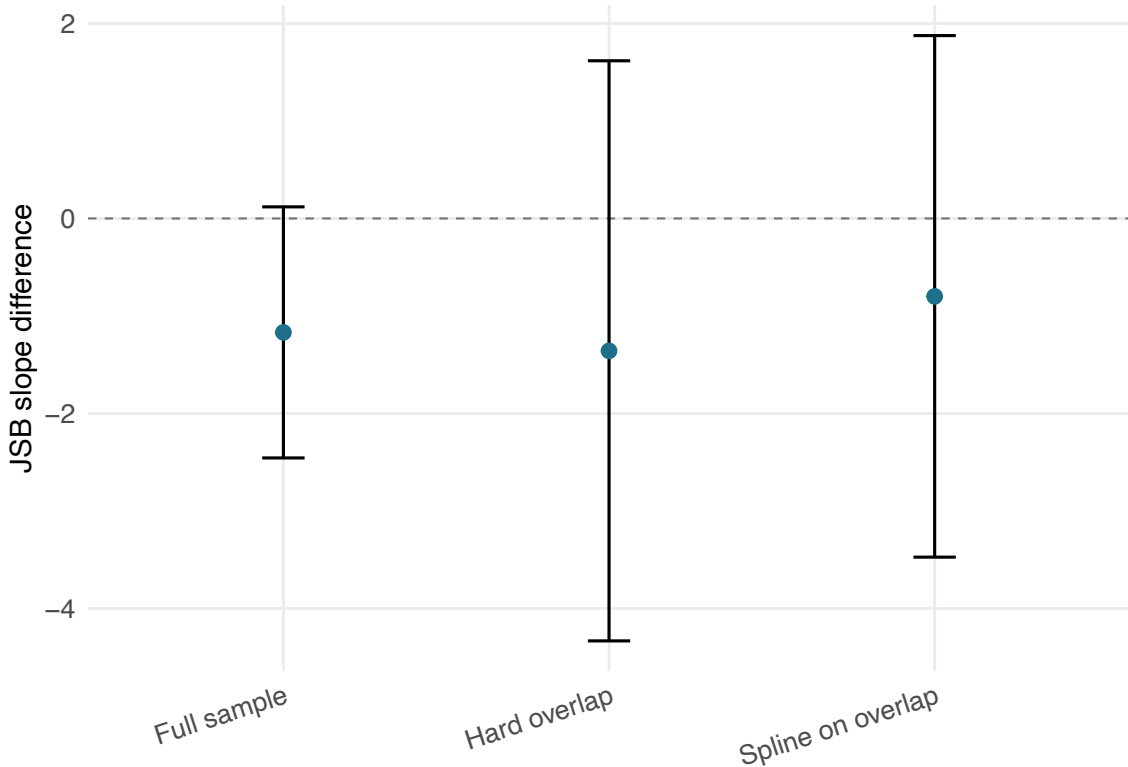


Figure E.1: Full-sample and common-support JSB slope differences

Notes: Points reproduce the JSB regime differences in Table 3. Whiskers are pointwise CR2 95% intervals with contrast-specific Satterthwaite degrees of freedom. The spline uses the full estimation sample and is evaluated on overlap. The fragile exact-matching comparison appears in the table.

The unweighted endpoint-regime analogue excludes transition quarters. In this sample it contains 684 observations and yields a JSB slope difference of -0.856 , with a clustered p -value of 0.025 and a CR2 p -value of 0.080. This estimate and the hard-support and spline diagnostics address different support populations and are estimated on different samples, so they should not be read as competing estimates of one quantity; exact endpoint matching is fragile in the sense just described.

F Timing and Placebo Diagnostics

The endpoint grid re-estimates the historical specification through successive fourth-quarter endpoints. The estimates are positive at the earliest effective endpoints, turn negative by 2019Q4, and become more negative as later observations enter the sample. The pseudo-event exercise places analogous breaks entirely before the first direct deposit-ceiling change.

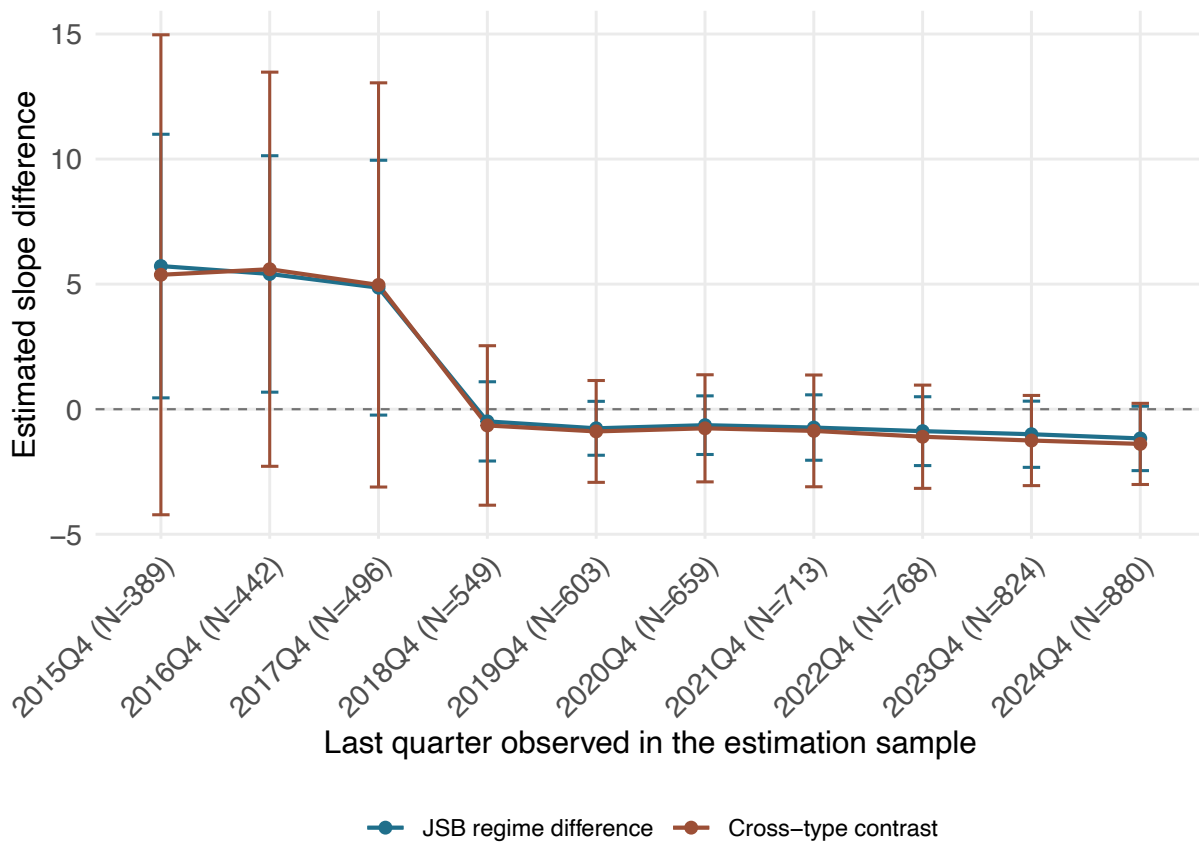


Figure F.1: Cumulative endpoint estimates

Notes: Each point uses a cumulative sample through the labeled endpoint, with observation count on the axis. Whiskers are CR2 95% intervals with Satterthwaite degrees of freedom. The 2013Q4 and 2014Q4 gradient-difference contrasts fail the design-matrix row-space test and are omitted. Later displayed contrasts are estimable. Changes between points reflect added observations, not event-time responses or a dated structural break.

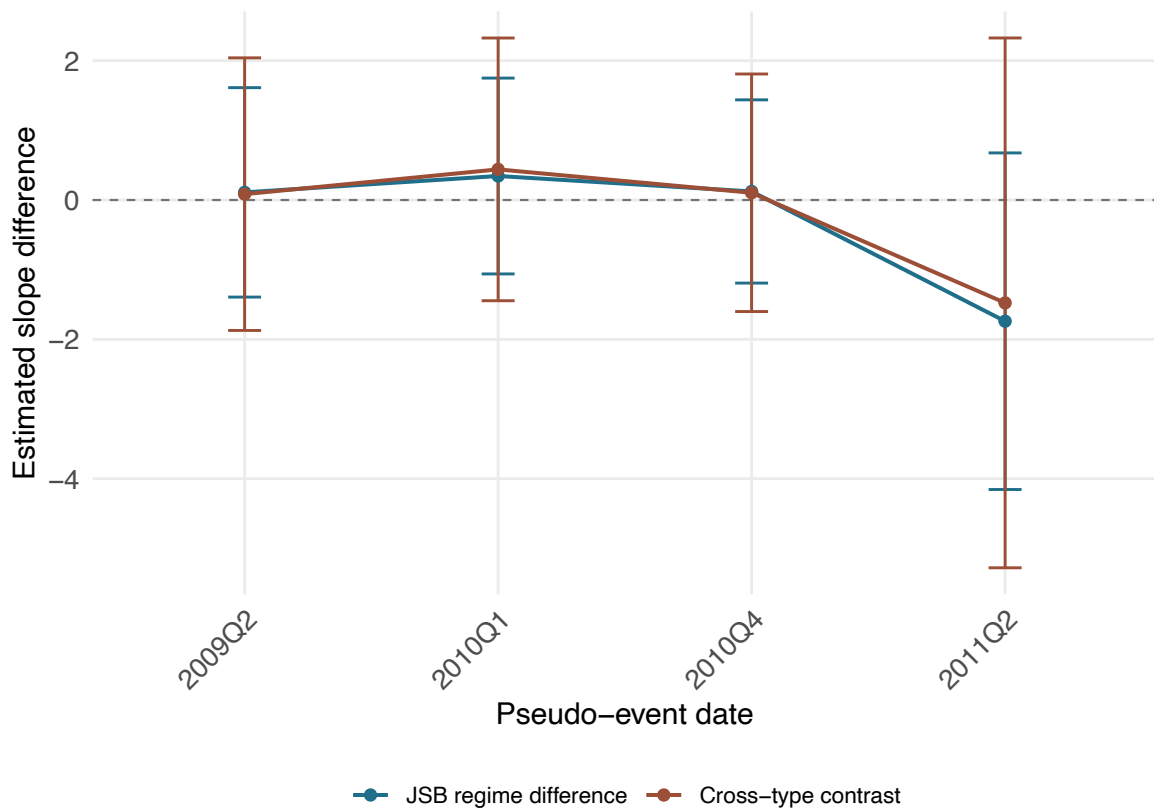


Figure F.2: Pre-reform functional-form placebo estimates

Notes: Pseudo-break regressions use observations before 2012Q2, with thirteen banks. Whiskers are conventional bank-clustered 95% intervals with twelve degrees of freedom. The four dates are retained, including the imprecise negative estimate at 2011Q2. These are functional-form diagnostics, not randomization tests.

The flexible-path specification is

$$Y_{it} = \alpha_i + \lambda_{g(i),t} + \sum_g \sum_{q \neq q_0} a_{gq} \mathbf{1}\{g(i) = g, t = q\} E_i + u_{it}, \quad q_0 = 2011Q3. \quad (\text{F.1})$$

The raw path is $a_t = a_{Jt} - a_{Gt}$. The adjusted path subtracts $b_t = \hat{\beta}_{1J}(RRR_{Jt} - RRR_{Jq_0}) - \hat{\beta}_{1G}(RRR_{Gt} - RRR_{Gq_0})$. It is normalized to zero in q_0 and describes exposure-gradient movement rather than a causal event response.

Let X_f, X_m denote full-rank representations of the flexible and main designs, e_f, e_m their residuals, and l_{ft}, l_{mt} the corresponding contrast vectors. For bank i , the mapped influence difference is

$$s_{it} = l'_{ft}(X'_f X_f)^{-1} X'_{fi} e_{fi} - l'_{mt}(X'_m X_m)^{-1} X'_{mi} e_{mi}, \quad \hat{\Sigma}_{tu} = \frac{G}{G-1} \sum_i s_{it} s_{iu}, \quad (\text{F.2})$$

where $G = 14$. This stacked cluster sandwich includes cross-model and cross-quarter covariance, with the stated $G/(G-1)$ normalization and no additional residual-degrees-of-freedom factor. Point-wise bands use a 90 percent normal multiplier. The pre-period trend uses deterministic calendar weights $A_t = (t - \bar{t}) / \sum_q (q - \bar{t})^2$ over pre-2012Q2 quarters excluding q_0 , variance $A \hat{\Sigma} A'$, and a t_{13} reference. It does not replace the full covariance by its diagonal or treat the fitted baseline as known.

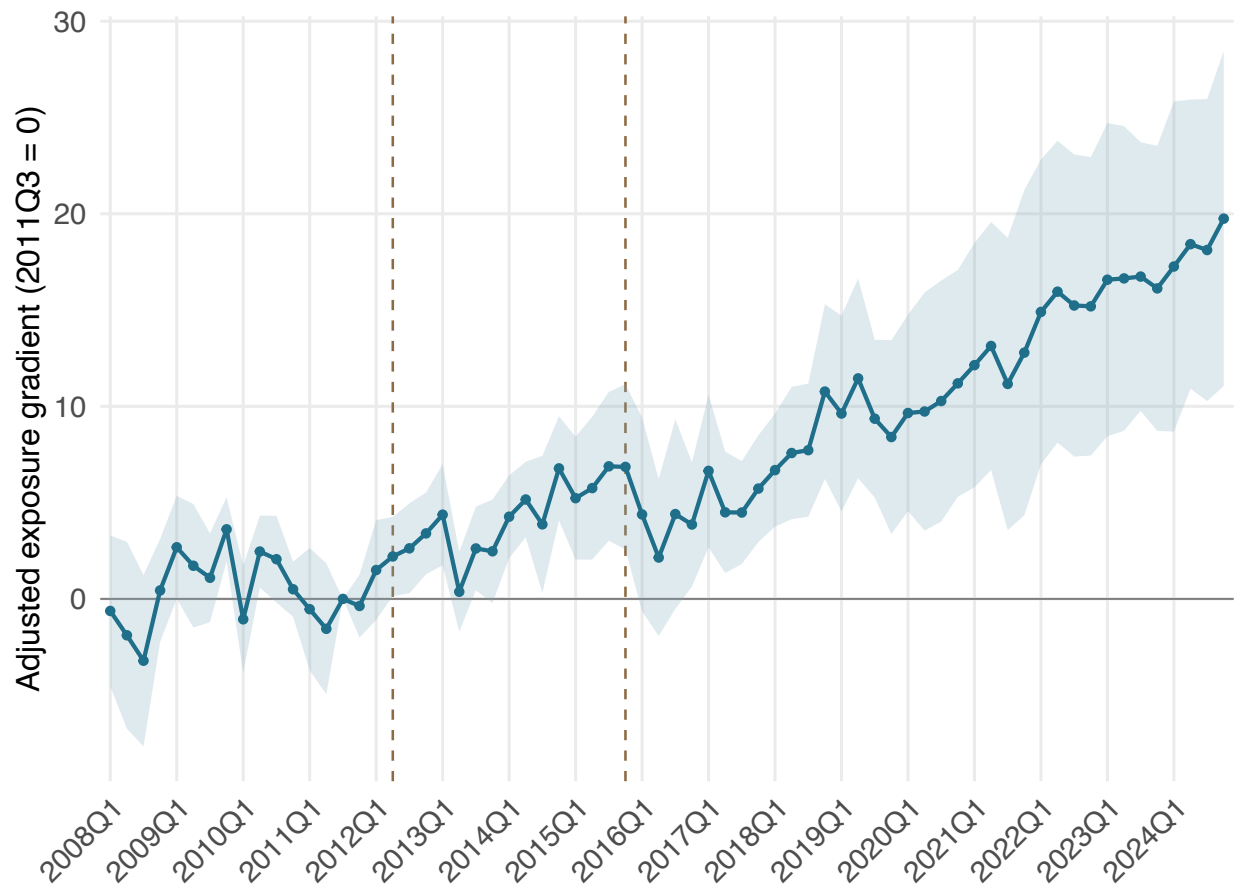


Figure F.3: Baseline-adjusted exposure-gradient path for market funding

Notes: The JSB-minus-large flexible path in Eq. (F.1), less the fitted regulated-period RRR component, is normalized to 2011Q3. Shading denotes pointwise 90% normal intervals using the joint bank-clustered covariance in Eq. (F.2); these are not simultaneous bands. Vertical lines mark 2012Q2 and 2015Q4. The trend test uses the full cross-quarter covariance and a t_{13} reference.

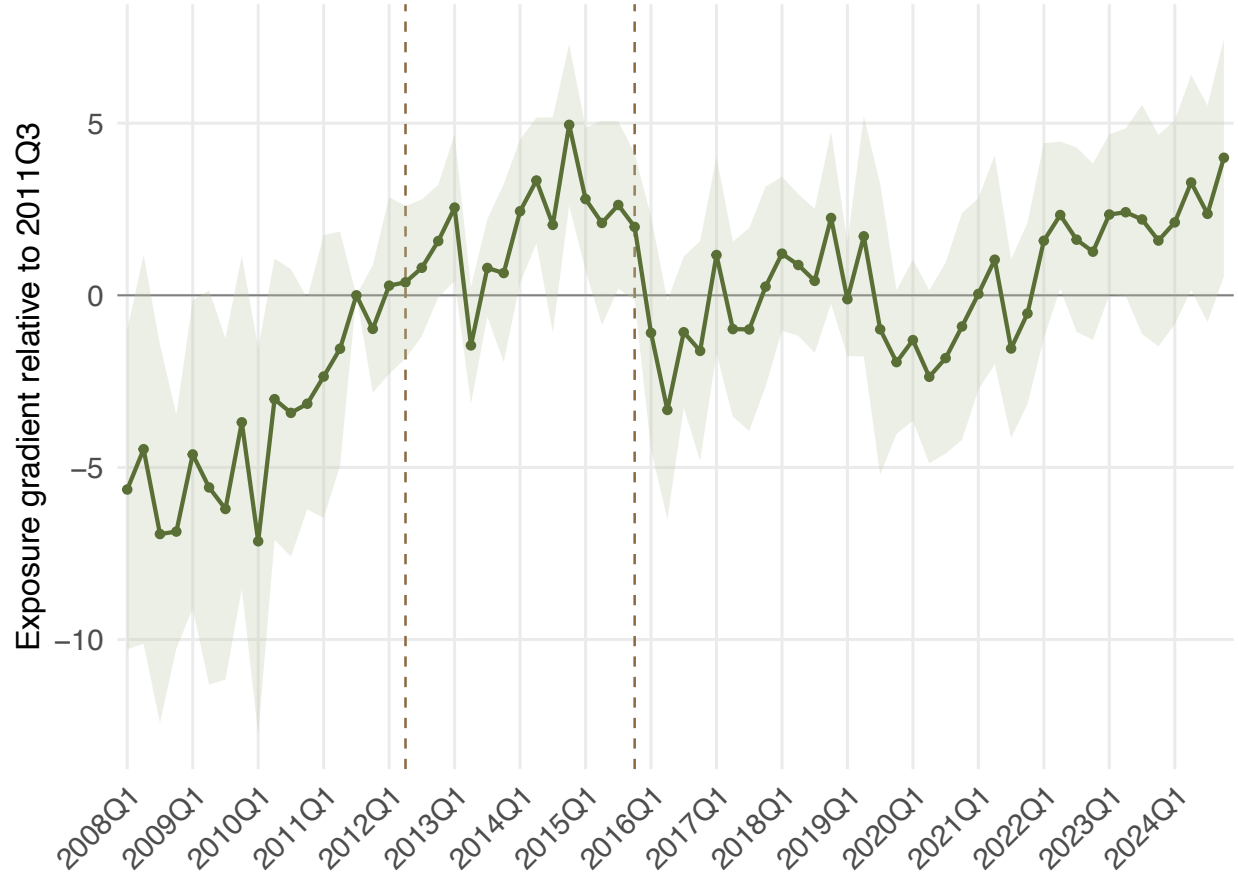


Figure F.4: Raw quarter-specific exposure-gradient path for market funding
Notes: The figure shows $a_{Jt} - a_{Gt}$ from Eq. (F.1), normalized to 2011Q3, on the 880-observation sample. Shading denotes pointwise 90% normal intervals using the flexible-model bank-clustered sandwich with factor $G/(G - 1)$, $G = 14$. Vertical lines mark 2012Q2 and 2015Q4. The path is a functional-form diagnostic, not a causal event response.

Table F.1: Flexible exposure-path diagnostics

Statistic	Value
Raw pre-period equality test	Not computable
Raw linear pre-trend p -value	0.003
Baseline-adjusted residual-trend slope	0.064
Baseline-adjusted residual-trend p -value	0.423
Maximum post-reform single-quarter share	0.136
Omitted quarter	2011Q3

Notes: Tests refer to the joint-stock-minus-large-bank flexible market-funding path. Raw coefficients are differences relative to the omitted quarter. The joint pre-period equality test imposes sixteen restrictions, but the bank-clustered covariance matrix of those sixteen linear combinations has rank eleven with fourteen clusters, so the Wald statistic is not computable under bank-clustered inference and no F -statistic is reported. The linear pre-trend test uses a single contrast and is unaffected by that rank deficiency. The baseline-adjusted residual-trend test subtracts the fitted regulated-period RRR-exposure component before testing residual pre-period movement. Its variance is $A\Sigma A'$, where A contains the calendar-trend weights and Σ is the full joint bank-clustered covariance in Eq. (F.2). The adjusted trend uses a t_{13} reference. The single-quarter share is the largest absolute quarter-to-quarter movement in the adjusted path from 2015Q4 onward as a fraction of the summed absolute movements over that period.

G Alternative Measurements and Bank Influence

Alternative regime indices change the coding and weighting of the deposit-pricing chronology without introducing a new cross-sectional source of policy variation. Alternative exposures use only information available through 2011Q4. The leave-one-bank-out and permutation exercises evaluate concentration in individual banks and in the observed within-type exposure assignment.

Table G.1: Alternative regime indices and RRR measures

Index RRR measure	Est.	Cluster p	CR2 p	Obs.
Equal-step pure ceiling index RRR_eoq	-1.388	0.009	0.071	880
Equal-step pure ceiling index RRR_avg	-1.425	0.014	0.080	880
Equal-step pure ceiling index RRR_boq	-1.385	0.016	0.086	880
Cap-and-scope index RRR_eoq	-1.362	0.010	0.072	880
Cap-and-scope index RRR_avg	-1.397	0.014	0.081	880
Cap-and-scope index RRR_boq	-1.354	0.017	0.087	880

Notes: Rows report the JSB-minus-large slope-change contrast for the market-funding outcome under each combination of deposit-pricing index and RRR timing convention.

Table G.2: Alternative predetermined exposure definitions

Exposure definition	Contrast	Cluster p	CR2 p	Obs.	Banks
2008Q1–2011Q4 baseline	-1.388	0.009	0.071	880	14
2009Q1–2011Q4	-1.472	0.004	0.056	880	14
2010Q1–2011Q4	-1.573	0.003	0.035	880	14
2011Q1–2011Q4	-1.399	0.003	0.038	880	14
Last pre-reform observation	-1.306	0.002	0.057	880	14
Within-type standardized baseline	-0.811	0.013	0.045	880	14
Size-residualized baseline	-0.660	0.006	0.058	849	13
Size- and type-residualized baseline	-0.694	0.005	0.038	849	13

Notes: All rows report the cross-type market-funding gradient difference using pre-2012 information. Window and last-report rows use the fixed baseline mean and standard deviation. Within-type standardization subtracts each type’s mean and divides by its own sample standard deviation across the fourteen banks, changing the relative scales of the two type coefficients. The final two rows regress baseline E_i on an intercept and pre-reform log assets, or those variables plus a bank-type indicator, then standardize the residuals across the thirteen complete banks. Zheshang lacks pre-reform log assets. Thus these rows change units and, for residualization, the sample; their coefficient range is not a common-unit magnitude comparison. On the same thirteen-bank sample the baseline common-scale contrast is -1.490 .

Table G.3: Leave-one-bank-out headline contrast

Dropped bank	Est.	SE	Cluster p	CR2 p	Obs.	Banks	Flip
Ping An	-1.650	0.535	0.009	0.078	817	13	No
Minsheng	-1.628	0.580	0.016	0.085	815	13	No
Bank of China	-1.527	0.447	0.005	0.113	813	13	No
Zheshang	-1.490	0.447	0.006	0.057	849	13	No
SPDB	-1.424	0.433	0.006	0.060	812	13	No
CITIC	-1.414	0.526	0.020	0.109	812	13	No
CCB	-1.399	0.462	0.010	0.087	814	13	No
ABC	-1.370	0.495	0.017	0.118	820	13	No
Industrial	-1.363	0.512	0.021	0.098	812	13	No
Everbright	-1.362	0.457	0.011	0.075	827	13	No
ICBC	-1.348	0.465	0.013	0.088	813	13	No
China Merchants	-1.313	0.490	0.020	0.096	812	13	No
BOCOM	-1.265	0.446	0.015	0.143	812	13	No
Hua Xia	-0.959	0.294	0.007	0.053	812	13	No

Notes: Each row re-estimates the primary pure-index national joint-stock-minus-large-bank contrast after dropping one bank. “Flip” indicates a sign change relative to the full-sample estimate.

Table G.4: Exposure-trend sensitivity

<i>A. Estimates (bank-clustered standard errors in parentheses)</i>				
Specification	JSB regime difference	Large regime difference	Cross-type contrast	CR2 p (contrast)
Baseline	-1.168 (0.435)	0.219 (0.134)	-1.388 (0.455)	0.071
Common exposure trend	0.728 (0.342)	2.186 (0.504)	-1.458 (0.474)	0.070
Type-specific exposure trends	1.011 (0.337)	-0.054 (0.632)	1.065 (0.717)	0.342
<i>B. Conditioning of the substantive regressors, fixed effects partialled out</i>				
Specification	Regressors	Exact dependence	Condition number	Max VIF
Baseline	6	No	17.31	71.3
Common exposure trend	7	No	33.62	202.1
Type-specific exposure trends	8	No	40.05	314.0

Notes: Panel A reports all three contrasts so that the within-type regime differences, not only the cross-type contrast, can be read off. **A common exposure trend leaves the cross-type contrast similar but changes both within-type regime differences substantially, reversing the joint-stock difference while the large-bank difference stays positive and rises. Allowing exposure trends to differ by bank type also reverses the cross-type point estimate.** The baseline pattern is therefore sensitive to the modeling of exposure-related trends. Panel B diagnoses conditioning after partialling the bank and bank-type-by-quarter fixed effects out of the substantive regressors and standardizing the residualized columns, which is the relevant comparison: the condition number of the fixed-effect-inclusive design matrix exceeds 10^{16} in every specification, including the baseline, and reflects the dummy parameterization rather than the added economic variables. **No specification exhibits exact linear dependence among the substantive regressors**, so the type-specific trend result is a genuine sensitivity of the point estimate and not an artifact of a rank-deficient design. Replacing calendar time with centred and scaled time leaves every contrast unchanged to within 10^{-14} .

Table G.5: Within-type exposure permutation sensitivity

Statistic	Value
Observed JSB-minus-large contrast	-1.388
Permutation draws	999
Permutation tail proportion	0.012

Notes: The bank-level exposure vector is randomly permuted within each bank type, without replacement within a draw, holding outcomes, policy paths, samples, and fixed effects fixed. There are 999 Monte Carlo reassignments, with seed 20260812. The reported tail proportion counts $|\hat{\delta}^*| \geq |\hat{\delta}|$ (ties included), divided by 999; no plus-one correction is applied. All draws return estimates. Because actual exposure is not randomly assigned, this is a sensitivity diagnostic, not design-based randomization inference.

Table G.6: Fitted gradient magnitudes

<i>Panel A. One-standard-deviation exposure gap, one percentage point higher RRR level</i>			
Deposit-pricing regime	Fitted gap difference	CR2 95% CI	
Regulated	1.144	[-0.330, 2.618]	
Post-ceiling-removal	-0.024	[-0.453, 0.404]	
Historical regime-associated difference	-1.168	[-2.456, 0.119]	
<i>Panel B. Interquartile exposure gap (0.910 standard-deviation units), one percentage point lower RRR level</i>			
Deposit-pricing regime	Fitted gap difference	CR2 95% CI	
Regulated	-1.041	[-2.382, 0.300]	
Post-ceiling-removal	0.022	[-0.368, 0.412]	
<i>Panel C. One-standard-deviation exposure gap, representative RRR level differences</i>			
RRR difference	Regulated	Post-ceiling	Difference
50 basis points lower	-0.572	0.012	0.584
100 basis points lower	-1.144	0.024	1.168

Notes: The table evaluates the estimated levels gradients at stated exposure gaps and RRR levels. Each entry is a difference between fitted market-funding shares, in percentage points of liabilities, for two joint-stock banks that differ in predetermined exposure and are compared at RRR levels the stated distance apart. A negative entry in Panels B and C means the fitted gap between the higher- and lower-exposure bank is smaller at the lower RRR level. These are comparisons of fitted values across the observed cross-section, not estimates of how any bank adjusts after the central bank changes the ratio. Panel B uses a P75–P25 exposure gap of 0.910 standard-deviation units across the nine joint-stock banks. CR2 intervals scale the corresponding model-based CR2 confidence intervals. These calculations are arithmetic on estimated levels coefficients; they are not structural counterfactuals, treatment effects, or dynamic responses.

H Liability Definitions and Accounting

Market funding is bonds payable plus repo liabilities. Noncustomer funding is bonds payable plus reported interbank liabilities, which already include repo liabilities; it is not the sum of market

funding and interbank liabilities, which would count repo twice. Component outcomes are reported to show whether the historical aggregate pattern is concentrated in one reported liability category. Because component coverage differs across outcomes, these regressions are descriptive mechanism diagnostics rather than a decomposition of the full-sample coefficient.

Table H.1: Liability-component estimates

Outcome	JSB difference	Cluster p	CR2 p	Cross-type	CR2 p	Obs.
Customer deposits / liabilities	-0.004	0.995	0.996	1.078	0.360	892
Market funding (bonds + repo) / liabilities	-1.168	0.019	0.066	-1.388	0.071	880
Noncustomer funding, each component once / liabilities	-0.205	0.551	0.602	-1.766	0.397	889
Interbank liabilities (includes repo) / liabilities	0.176	0.512	0.566	-1.088	0.487	892
Repo liabilities / liabilities	-0.788	0.084	0.153	-0.695	0.292	882
Residual liabilities / liabilities	0.213	0.598	0.646	0.708	0.589	889
Bonds payable net of NCD, excluding BOCOM / liabilities	0.589	0.235	0.279	0.913	0.279	571

Notes: Each row re-estimates the primary specification with the stated liability share as the outcome, reporting the joint-stock regime-associated slope difference and the joint-stock-minus-large-bank contrast. **Unlike Table 4, rows here use their own available samples**, whose observation counts are the final column; the main-text table restricts every outcome to the same baseline observations. The rows overlap and must not be summed: reported interbank liabilities already include repo liabilities, so repo enters both the market-funding measure and the interbank measure, and the noncustomer row counts each component once rather than summing the two. The accounting partition of liabilities is reported in Table 4. The final row subtracts reconstructed NCD outstanding from bonds payable; that subtraction is valid only for banks whose accounts place NCDs inside bonds payable, which fails for Bank of Communications, whose 23 negative cells are excluded with the bank.

I NCD Access, Adoption, and Usage

The security-level input combines Shanghai Clearing House issuance notices, collected through the documented public-source crosswalk, with bank identifiers. Successful issues require positive issued amounts and valid value and maturity dates. The amount is the actual issue amount where observed, otherwise the clearing notice’s total amount. Gross issuance is assigned by value (settlement) date; quarter-end outstanding includes issues with value date at or before quarter-end and maturity strictly after quarter-end. Net issuance is gross issuance less the amount maturing in that quarter. Issuance announcement dates identify first observed issuance, not the stock’s settlement date.

Let B_{it} be outstanding NCDs, I_{it} gross issuance, and T_{it} current total liabilities. Define $U_{it} = 100B_{it}/T_{it}$ and $Q_{it} = 100I_{it}/T_{it}$. The stock-change outcome is $U_{i,t+2} - U_{it}$; the cumulative-flow outcome is $Q_{it} + Q_{i,t+1} + Q_{i,t+2}$, with each quarter’s own denominator. Lags and leads require the stated calendar quarters, including missing values when a required quarter is absent.

The legal-access indicator A_{it} starts in the first full quarter after the documented eligibility date. The access sample includes only the nine banks with documented December 2013 eligibility (evidence grade B); unknown eligibility is not coded as ineligibility. The first-issuance indicator F_{it} starts in the first full quarter after the first observed issue. For each retained layer, the augmented

equation is

$$Y_{it} = \text{right-hand side of Eq. (5)} + \gamma RRR_{g(i),t} E_i Z_{it}, \quad Z_{it} \in \{A_{it}, F_{it}, U_{i,t-1}\}. \quad (\text{I.1})$$

Each regression adds one pooled interaction, not a bank-type-specific NCD term or additional level control. Each baseline/augmentation pair uses identical complete observations. Access changes the feasible set; adoption and usage remain endogenous funding choices. Direct-usage outcomes use Eq. (5) with the relevant quantity as Y .

Subtracting reconstructed NCD balances from bonds payable additionally requires comparable financial-statement classification. The source reconciliation flags Bank of Communications for placement concerns; its exclusion applies only to that exploratory residual-bond calculation, not the headline market-funding sample. Negative residual amounts alone do not establish a particular accounting classification, and this residual-bond calculation is not used for the paper’s main conclusions.

Table I.1: NCD access, adoption, and realized usage

Model or outcome	Estimate	Cluster p	CR2 p	Obs.	Banks
<i>Legal access</i>					
Matched baseline	-1.636	0.017	0.165	600	9
Augmented contrast	-1.625	0.018	0.169	600	9
legal-access interaction	0.091	0.312	0.430	600	9
<i>First issuance</i>					
Matched baseline	-1.388	0.009	0.071	880	14
Augmented contrast	-1.397	0.009	0.071	880	14
First-issuance interaction	0.123	0.085	0.108	880	14
<i>Lagged usage</i>					
Matched baseline	2.200	0.390	0.491	587	14
Augmented contrast	4.741	0.063	0.192	587	14
lagged NCD usage interaction	0.055	0.010	0.040	587	14
<i>Direct usage outcome</i>					
NCD outstanding / liabilities	3.360	0.122	0.238	617	14
NCD outstanding share change, t to $t + 2$	-3.214	0.027	0.124	586	14
Gross NCD issuance / liabilities	1.847	0.195	0.310	617	14
Gross NCD issuance shares, summed t to $t + 2$	0.683	0.878	0.896	578	14

Notes: The first three panels report, for each NCD object, the headline contrast on the matched sample, the contrast after adding the corresponding NCD interaction, and the interaction coefficient itself. The direct-usage rows replace the outcome with a realized NCD quantity and are estimated on every bank-quarter for which that quantity is observed; they are post-adoption funding choices and are reported as descriptive funding evidence, not as controls.

J Annual Deposit-Side Evidence

This appendix examines reported deposit rates and deposit composition on an unbalanced common annual sample. Rate disclosures are full-year averages, balance-sheet stocks are Q4 values, and the RRR and regime interactions are calendar-year averages of their quarterly counterparts. The sample is the set of bank-years in which the annual report discloses all four deposit-rate components and the fourth-quarter market-funding outcome is observed. It covers ten banks between 2008 and 2024 and contains 156 bank-year observations. All outcomes in the annual comparison use the same observations.

For annual outcome Y_{iy} the design averages each quarterly interaction:

$$Y_{iy} = \alpha_i + \lambda_{g(i),y} + \sum_g \mathbf{1}\{g(i) = g\} [\beta_{1g} \overline{RRRE}_{iy} + \beta_{2g} \overline{LE}_{iy} + \beta_{3g} \overline{RRLE}_{iy}] + u_{iy}. \quad (\text{J.1})$$

The bar averages observed calendar-year quarterly design terms, preserving missing inputs rather than interpolating them; it does not replace $\overline{RRRL}$ by $\overline{RRR} \overline{L}$. The four rate inputs are corporate demand, corporate time, personal demand, and personal time deposit rates, each the full-year average reported in the annual report. Stocks use Q4. Inference clusters by bank. Annual NCD augmentations add the annual mean of the corresponding quarterly interaction in Eq. (I.1), with a common complete-case sample for each pair. The composite reported deposit-rate index is the equal-weight mean of corporate and personal demand- and time-deposit rates; it is not a balance-weighted effective deposit cost. These estimates characterize associated liability quantities, composition, and reported rates and do not constitute mediation evidence.

Table J.1: Annual deposit-side sample

Bank	Type	First year	Last year	Bank-years
Agricultural Bank of China	Large	2008	2024	17
Bank of China	Large	2008	2024	17
Bank of Communications	Large	2020	2024	5
China Construction Bank	Large	2008	2024	16
Industrial and Commercial Bank of China	Large	2008	2024	17
China CITIC Bank	JSB	2008	2024	17
China Everbright Bank	JSB	2008	2024	17
China Merchants Bank	JSB	2008	2024	17
China Minsheng Banking Corp.	JSB	2009	2024	16
Shanghai Pudong Development Bank	JSB	2008	2024	17
Total		2008	2024	156

Notes: The sample requires all four annual rate components and a Q4 market-funding outcome; it contains 156 bank-years for ten banks. Bank of Communications rate disclosures begin in 2020. Annual construction follows Eq. (J.1).

Table J.2: Annual liability quantities

Outcome	JSB pre	JSB change	JSB full	Large change	JSB-large	CR2 p
Market funding	0.025	-0.852	-0.827	0.241	-1.093	0.429
Bonds payable	0.123	-0.957	-0.834	0.310	-1.267	0.243
Repos	-0.098	0.105	0.007	-0.069	0.174	0.842
Deposits / liabilities	0.103	0.856	0.959	-0.784	1.640	0.462
Deposits / assets	0.288	0.495	0.783	-1.170	1.664	0.427
Time-deposit share	2.659	-2.430	0.229	1.079	-3.510	0.073

Notes: All rows use the common 156 bank-years and Eq. (J.1). Stocks are Q4 values. Market funding, bonds, repos, and deposits/liabilities are percentage shares of liabilities; deposits/assets and the time-deposit share use their labeled denominators. CR2 p -values use bank clustering and Satterthwaite degrees of freedom for the cross-type contrast. Bonds plus repos equal market funding on the same design and sample.

Table J.3: Annual reported deposit rates

Outcome	JSB pre	JSB change	JSB full	Large change	JSB-large	CR2 p
Composite rate index	0.025	-0.030	-0.004	0.025	-0.055	0.462
Demand-deposit rate index	-0.003	0.002	-0.002	0.021	-0.019	0.485
Time-deposit rate index	0.054	-0.061	-0.007	0.029	-0.090	0.502
Corporate-deposit rate index	0.008	-0.017	-0.010	0.073	-0.091	0.140
Personal-deposit rate index	0.043	-0.042	0.001	-0.023	-0.018	0.846

Notes: All rows use the common 156 bank-years and Eq. (J.1). Rates are reported full-year averages in percentage points. The composite equally weights the four corporate/personal demand/time rates; the remaining indices equally weight the indicated pair. They are not balance-weighted effective funding costs. CR2 p -values concern the cross-type contrast.

Table J.4: Annual three-layer NCD evidence

NCD layer	Outcome	Baseline	CR2 p	Augmented	CR2 p	Obs.	Banks
Legal access	Market funding	-1.118	0.693	-1.115	0.694	123	8
Legal access	Bonds payable	-1.637	0.338	-1.636	0.338	123	8
Legal access	Deposits / liabilities	3.015	0.472	3.010	0.473	123	8
Legal access	Composite rate index	-0.090	0.335	-0.090	0.335	123	8
First issuance	Market funding	-1.093	0.429	-1.072	0.440	156	10
First issuance	Bonds payable	-1.267	0.243	-1.259	0.247	156	10
First issuance	Deposits / liabilities	1.640	0.462	1.593	0.471	156	10
First issuance	Composite rate index	-0.055	0.462	-0.055	0.457	156	10
Lagged usage	Market funding	-7.073	0.360	-1.394	0.840	104	10
Lagged usage	Bonds payable	-0.926	0.849	4.839	0.388	104	10
Lagged usage	Deposits / liabilities	7.416	0.471	1.307	0.909	104	10
Lagged usage	Composite rate index	-0.042	0.799	0.056	0.725	104	10

Notes: Each pair uses identical complete bank-years, with and without the annual mean of the NCD interaction in Eq. (I.1). CR2 p -values concern the cross-type contrast. Legal access uses documented December 2013 eligibility; unknown eligibility is excluded. Lagged usage requires the preceding calendar quarter.

Table J.5: Annual robustness

Check	Outcome	Estimate or range	CR2 p	Sample
Exposure windows	Market funding	[-1.120, -0.961]	—	5 windows
Exposure windows	Composite rate index	[-0.055, -0.030]	—	5 windows
Cap/scope index	Market funding	-1.087	0.433	156 obs.
Cap/scope index	Composite rate index	-0.055	0.460	156 obs.
Leave-one-bank-out	Market funding	[-2.048, 0.095]	—	9-bank runs
Leave-one-bank-out	Composite rate index	[-0.152, 0.001]	—	9-bank runs

Notes: All estimates are annual cross-type contrasts. The exposure-window rows use the fixed baseline scale. Within-type standardization and residualization change units and are shown separately in the measurement audit; their coefficients are not pooled into a common-unit range. The market-funding leave-one-bank-out range crosses zero. Other conventions follow Eq. (J.1).